

ARTIFICIAL INTELLIGENCE AND HUMAN LEGAL REASONING

Nick Bednar,^{*} David Cleveland,^{**} Allan Erbsen,^{***}
and Daniel Schwarcz^{****}

Abstract

Empirical evidence increasingly demonstrates that generative artificial intelligence has the capacity to improve the speed and quality of legal work, yet many lawyers, judges, and clients are reluctant to fully embrace AI. One important reason for hesitation is the concern that AI may undermine the human reasoning and judgment on which competent legal practice depends. This Article provides the first empirical evidence evaluating that concern by testing whether upper level law students who rely on AI at an early stage of a project experience reduced comprehension and impaired legal reasoning on later stages when AI is not an available option.

To evaluate the possibility that AI degrades comprehension and reasoning, we conducted a randomized controlled trial involving approximately one hundred second and third year law students at the University of Minnesota Law School. Participants completed four sequential lawyering tasks: writing a memo synthesizing the law based on a packet of legal materials, answering closed-book multiple choice questions that tested their comprehension of the materials, writing a memo applying the materials to a fact pattern, and revising their second memo. Participants were randomly assigned either to a control group, which could not use AI until the final revision task, or to an AI-exposed group, which used AI during both the initial synthesis task and the final revision task, but not during the intervening comprehension and application tasks.

The results provide a more complex picture of AI's effects on legal reasoning than critics or enthusiasts often assume. As expected, participants who used AI to help craft synthesis memos produced substantially stronger work and completed that task more quickly. But contrary to our preregistered hypothesis, AI exposure at this initial stage did not diminish downstream

^{*} McKnight Land-Grant Professor, University of Minnesota Law School.

^{**} Clinical Professor of Law and Director of Legal Research & Writing, University of Minnesota Law School.

^{***} Popham, Haik, Schnobrich/Lindquist & Venum Professor of Law, University of Minnesota Law School.

^{****} Fredrikson & Byron Professor of Law, University of Minnesota Law School. For helpful comments and suggestions, we thank Matthew Bodie, William Childs, Kevin Frazier, Chad Ratashak, and participants at the University of Minnesota Law School Faculty Works in Progress Workshop.

comprehension of the underlying legal principles. To the contrary, participants who used AI on the synthesis task outperformed the control group on the later application task even when neither group had access to AI. Yet when all participants used AI to revise their reasoning memos, participants who started with weaker memos improved while participants who started with stronger memos regressed. These findings suggest that AI does not inevitably erode or promote independent legal reasoning, but that its effects depend on when and how law students and junior lawyers use AI. The Article builds on this insight by suggesting best practices for AI use and avenues for further empirical research.

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Introduction

Since the public release of ChatGPT in 2022, empirical evidence has increasingly suggested that generative AI systems are capable of fundamentally transforming the practice of law, particularly with respect to work traditionally performed by junior lawyers and attorneys operating in less complex fields. This evidence includes studies showing that AI systems can earn top marks on legal exams,¹ that upper-level law students who use AI

¹ See Jonathan H. Choi, Kristin E. Hickman, Amy B. Monahan & Daniel Schwarcz, *ChatGPT Goes to Law School*, 71 J. LEGAL EDUC. 387, 388–91 (2022); Daniel Martin Katz, Michael James Bommarito, Shang Gao & Pablo Arredondo, *GPT-4 Passes the Bar Exam*,

tools consistently produce higher-quality legal work in less time than those who do not,² and that these advantages persist in real-world field experiments.³

Despite this mounting evidence, many actors across the legal ecosystem remain reluctant to fully embrace AI.⁴ Although the number of lawyers, law firms, and clients that categorically reject AI use is rapidly diminishing, it is still substantial.⁵ Even among lawyers who incorporate AI into their work, most continue to treat it as a limited tool, rather than as a transformative technology that fundamentally alters how they practice law.⁶ Consistent with these attitudes, there is little evidence that the emergence of AI has triggered fundamental shifts in the structure or economics of the legal profession, at least as of early 2026.⁷

PHIL. TRANS. R. SOC. A, Apr. 15, 2024, at 1; Andrew Blair-Stanek et al., GPT Gets Its First Law School B-pluses (Feb. 5, 2025) (unpublished manuscript), <https://ssrn.com/abstract=4717411>; cf. Eric Martínez, *Re-Evaluating GPT-4's Bar Exam Performance*, 32 A.I. & L. 1 (2024).

² See Jonathan H. Choi, Amy B. Monahan & Daniel Schwarcz, *Lawyering in the Age of Artificial Intelligence*, 109 MINN. L. REV. 147 (2024); Daniel Schwarcz, Sam Manning, JJ Prescott, Patrick J. Barry, David R. Cleveland, J.J. Prescott & Beverly Rich, *AI-Powered Lawyering: AI Reasoning Models, Retrieval Augmented Generation, and the Future of Legal Practice*, J.L. & EMPIRICAL ANALYSIS (forthcoming 2026), <https://doi.org/10.2139/ssrn.5162111>; Aileen Nielsen et al., *Building a Better Lawyer: Experimental Evidence That Artificial Intelligence Can Increase Legal Work Efficiency*, 21 J. EMPIRICAL LEGAL STUD. 979 (2024).

³ See Colleen V. Chien & Miriam Kim, *Generative AI and Legal Aid: Results from a Field Study and 100 Use Cases to Bridge the Access to Justice Gap*, 57 LOY. L.A. L. REV. 903 (2025). Although systematic evidence from practicing attorneys is less available for a variety of reasons, a growing number of practicing lawyers report that their practices have been fundamentally transformed by their embrace of AI. See, e.g., Zack Shapiro, *The Claude-Native Law Firm*, LINKEDIN (Feb. 27, 2026), <https://www.linkedin.com/pulse/claude-native-law-firm-zack-shapiro-qf7se>. At the same time, these anecdotal claims are not necessarily reliable.

⁴ See AM. BAR ASS'N, 2024 LEGAL TECHNOLOGY SURVEY REPORT (2024); Robert J. Ambrogi, *ABA Tech Survey Finds Growing Adoption of AI in Legal Practice, with Efficiency Gains as Primary Driver*, LOWNEXT (Mar. 20, 2025), <https://www.lawnext.com/2025/03/aba-tech-survey-finds-growing-adoption-of-ai-in-legal-practice-with-efficiency-gains-as-primary-driver.html>.

⁵ See *supra* note 4.

⁶ See Robert J. Ambrogi, *An AI-Assisted Look at Four New Surveys on AI Adoption in Law: How Do They Compare? Differ?*, LOWNEXT (Apr. 17, 2025), <https://www.lawnext.com/2025/04/an-ai-assisted-look-at-four-new-surveys-on-ai-adoption-in-law-how-do-they-compare-differ.html>.

⁷ See Daniel Schwarcz, Debarati Das, Dongyeop Kang & Brett H. McDonnell, *Thinking Like a Lawyer in the Age of Generative AI: Cognitive Limits on AI Adoption Among Lawyers*, J. INSTITUTIONAL & THEORETICAL ECON. (forthcoming 2026),

Various social, political, regulatory, and technical factors help explain the apparent gap between AI's increasingly impressive technical capacity and its muted practical impact on the legal profession.⁸ One of the most important is the belief among many lawyers that the use of AI in legal practice entails significant risks. These perceived risks fall into three categories: that AI will hallucinate source material,⁹ that using AI will jeopardize confidentiality,¹⁰ and that reliance on AI will erode human legal reasoning and professional judgment.¹¹

Although the legal profession has developed a range of strategies for managing hallucination and confidentiality concerns,¹² the possibility that AI use may erode human legal reasoning presents distinctive challenges that are not easily mitigated.¹³ Lawyers have long understood that sound judgment,

<https://ssrn.com/abstract=5260645>. But see Frank Landymore, *Law Firm Sacks Hundreds of Employees Amid Pivot to AI*, FUTURISM (Feb. 11, 2026, 10:25 AM EST), <https://futurism.com/artificial-intelligence/law-firm-sacks-hundreds-ai>; Maxim Massenkoff & Peter McCrory, *Labor Market Impacts of AI: A New Measure and Early Evidence*, ANTHROPIC 2 (March 5, 2026), <https://cdn.sanity.io/files/4zrzovbb/website/2b5bbaf2c1eb81dbf6e6fb813c1a24e35a64d376.pdf> (suggesting a large gap between AI's transformative potential and its actual adoption in the legal sector).

⁸ See Jennifer F. Reinganum, *On the Diffusion of New Technology: A Game Theoretic Approach*, 48 REV. ECON. STUD. 395 (1981).

⁹ See Matthew Dahl et al., *Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models*, 16 J. LEGAL ANALYSIS 64, 66 (2024); Varun Magesh et al., *Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools*, 22 J. EMPIRICAL LEGAL STUD. 216 (2025).

¹⁰ See Jennifer J. Cook & Denista R. Mavrova Heinrich, *AI-Ready Attorneys: Ethical Obligations and Privacy Considerations in the Age of Artificial Intelligence*, 72 U. KAN. L. REV. 313 (2023); John G. Browning, *Robot Lawyers Don't Have Disciplinary Hearings—Real Lawyers Do: The Ethical Risks and Responses in Using Generative Artificial Intelligence*, 40 GA. ST. U. L. REV. 917 (2023); The Insider & Thomas McKay, *Better Call Claude: The Confidentiality Risks of Getting Legal Advice from AI*, FORBES (Feb. 13, 2026, 5:02 PM EST), <https://www.forbes.com/sites/insider/2026/02/13/better-call-claude-the-confidentiality-risks-of-getting-legal-advice-from-ai>.

¹¹ See Schwarcz et al., *supra* note 7.

¹² See *infra* Part I.B.

¹³ See State Bar of Cal. Standing Comm. on Pro. Resp. & Conduct, *Practical Guidance for the Use of Generative Artificial Intelligence in the Practice of Law* 2–3 (Nov. 16, 2023), <https://www.calbar.ca.gov/Portals/0/documents/ethics/Generative-AI-Practical-Guidance.pdf> (warning that “[o]verreliance on AI tools is inconsistent with the active practice of law” and may “hinder[] critical attorney analysis fostered by traditional research and writing”); N.Y. City Bar Ass’n Comm. on Pro. Ethics, *Formal Op. 2024-5: Ethical Obligations of Lawyers and Law Firms Relating to the Use of Generative AI in the Practice of Law* (Aug. 7, 2024), https://www.nycbar.org/wp-content/uploads/2024/08/20221329_GenerativeAILawPractice.pdf (stating that use of

persuasive advocacy, and effective counseling depend on grappling directly with complex doctrine, messy facts, and difficult strategic tradeoffs.¹⁴ If AI tools mediate these processes, lawyers may reach answers more quickly but do so without internalizing the underlying reasoning.¹⁵ Attorneys with thinner conceptual foundations may overlook latent legal issues, offer short-term

generative AI “without the application of trained judgment ... is inconsistent with the competent and diligent practice of law” and cautioning against overreliance that “hinders critical attorney analysis”); Am. Bar Ass’n Standing Comm. on Ethics & Pro. Resp., Formal Op. 512: Generative Artificial Intelligence Tools (July 29, 2024), https://www.americanbar.org/content/dam/aba/administrative/professional_responsibility/ethics-opinions/aba-formal-opinion-512.pdf (explaining that lawyers’ “uncritical reliance on content” generated by AI can produce inaccurate advice and that lawyers may not “abdicate their responsibilities by relying solely” on such tools for tasks requiring professional judgment); Fla. Bar Ethics Op., Op. 24-1 (Jan. 19, 2024), <https://www.floridabar.org/etopinions/opinion-24-1> (cautioning that lawyers using generative AI “risk many of the same perils as those who have relied on inexperienced or overconfident nonlawyer assistants” and may not delegate to AI “any ... function that requires a lawyer’s personal judgment and participation”); N.C. State Bar, 2024 Formal Ethics Op. 1: Use of Artificial Intelligence in a Law Practice (2024), <https://www.ncbar.gov/for-lawyers/ethics/adopted-opinions/2024-formal-ethics-opinion-1> (requiring that a lawyer “must exercise her independent judgment” in supervising and deciding whether to rely on AI outputs); Council of Bars and L. Societies of Eur., CCBE Guide on the Use of Generative AI for Lawyers (Oct. 2, 2025), https://www.ccbe.eu/fileadmin/speciality_distribution/public/documents/IT_LAW/ITL_Guides_recommendations/EN_ITL_20251002_CCBE-guide-on-the-use-of-the-use-of-generative-AI-for-lawyers.pdf (noting that the “black box” nature of generative AI “makes it harder to verify the accuracy” of outputs and “may compromise the quality of legal advice”); N.Y. State Bar Ass’n Task Force on A.I., Report and Recommendations (Apr. 6, 2024), <https://nysba.org/wp-content/uploads/2022/03/2024-April-Report-and-Recommendations-of-the-Task-Force-on-Artificial-Intelligence.pdf> (warning of “possible skill erosion” such that “humans will no longer retain the knowledge to perform certain tasks”); D.C. Bar Legal Ethics Comm., Ethics Op. 388: Attorneys’ Use of Generative Artificial Intelligence in Client Matters (Apr. 11, 2024), <https://www.dcbar.org/for-lawyers/legal-ethics/ethics-opinions-210-present/ethics-opinion-388> (cautioning that lawyers who “blindly rely” on AI outputs act “at considerable peril”).

¹⁴ That understanding underlies the Socratic method in law school classrooms, the demanding hypotheticals judges pose at oral argument, and the many hours lawyers devote to meetings, depositions, and negotiations.

¹⁵ A simple hypothetical illustrates the risk. Imagine that a law school divides 1L students into two groups. Both attend the same classes, but they prepare differently. One group studies in the traditional way: reading the assigned materials, taking notes, and preparing a class outline. The second group does not do any of this. Instead, students in this group learn exclusively from AI summaries of sources and an AI-written course outline. Which group would perform better on their exams? We will never know because a university Institutional Review Board would rightly reject this experiment in pedagogical malpractice. But most will share the intuition that the students who trained entirely on AI-generated texts will not learn the reasoning skills that they need to succeed on exams and in practice.

advice that neglects downstream risks, pursue flawed strategies, struggle to explain legal principles to clients or courts, and miss important connections across doctrinal domains.¹⁶ Support for this concern comes from a growing body of empirical research in domains ranging from software development, to medical decision-making, to analytical writing, suggesting that individuals who depend heavily on AI tools often have weaker comprehension of the underlying material.¹⁷

For these reasons, determining when, how, and to what extent lawyers' use of AI affects human legal reasoning is critical to the future of legal practice and legal education. This Article provides the first such empirical evidence, reporting the results of a randomized controlled trial evaluating how AI use influences human legal reasoning. We recruited approximately 100 upper-level law students to participate in a study requiring them to complete four lawyering tasks in sequence:¹⁸

1. *Synthesis Task*: A writing task requiring participants to synthesize key legal principles from a set of complex source materials.
2. *Comprehension Task*: An objective assessment of participants' comprehension of synthesis task materials.
3. *Application Task*: A legal application task in which participants drafted an analytical memorandum applying the legal principles from the synthesis task to a hypothetical fact pattern.
4. *Revision Task*: A revision task requiring participants to improve their memorandum from the application task.

We randomly assigned participants to either a control group or an AI-exposed group. Participants in the control group could not use AI until the

¹⁶ See Schwarcz et al., *supra* note 7.

¹⁷ See Yizhou Fan et al., *Beware of Metacognitive Laziness: Effects of Generative Artificial Intelligence on Learning Motivation, Processes, and Performance*, 56 BRIT. J. EDUC. TECH. 489 (2025), <https://doi.org/10.1111/bjet.13544>; Nataliya Kosmyna et al., *Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task* (June 10, 2025) (unpublished manuscript), <https://arxiv.org/pdf/2506.08872>; Steven D. Shaw & Gideon Nave, *Thinking Fast, Slow, and Artificial: How AI Is Reshaping Human Reasoning and the Rise of Cognitive Surrender* (Jan. 11, 2026) (unpublished manuscript), <https://ssrn.com/abstract=6097646>. See Ahnaf Chowdhury Niloy et al., *Is ChatGPT a Menace for Creative Writing Ability? An Experiment*, 40 J. COMPUT. ASSISTED LEARNING 919 (2024), <https://doi.org/10.1111/jcal.12929>.

¹⁸ The experiment was submitted and approved by the University of Minnesota Institutional Review Board.

revision task. Participants in the AI-exposed group used AI during the synthesis and revision tasks, but not during the intervening comprehension and application tasks.

Consistent with the concern that AI usage can undermine human legal reasoning, we theorized that exposure to AI during initial legal tasks erodes human legal reasoning by reducing engagement with the underlying source materials. We thus hypothesized that while participants in the AI-exposed group would perform better on the synthesis task than those in the control group, they would perform worse on the comprehension and application tasks when AI was not available to either group.¹⁹

We find support for our first hypothesis that participants in the AI-exposed group would perform better than those in the control group on the synthesis task. Consistent with prior research, participants who used AI produced substantially stronger synthesis memos than participants who did not. In fact, AI availability increased performance on the synthesis task by roughly 50 to 70 percent, a magnitude that exceeds effects documented in earlier research using prior-generation AI models. Moreover, participants in the AI-exposed group completed the synthesis task significantly faster than participants in the control group.

Surprisingly, however, we find no evidence that the use of AI during the synthesis task degraded performance on either the comprehension or application tasks, even though AI was no longer available to either group for these tasks. We observe no statistically significant difference between the two groups on the comprehension task. And on the application task, we find the exact *opposite* of our hypothesized effect. Participants who used AI to complete the synthesis task outperformed the control group on the application task, even though neither group had access to AI when completing the application task.

What explains the AI-exposed group's superior performance on the application task? The data suggest that AI improved participants' ability to produce strong synthesis memos, which in turn had positive downstream effects on the quality of their reasoning memos even after AI was no longer available. In fact, the apparent effect of AI exposure on the reasoning memo

¹⁹ We pre-registered this hypothesis before conducting the experiment. See Nicholas R. Bednar, David Cleveland, Allan Erbsen & Daniel Schwarcz, *Legal Reasoning and Artificial Intelligence* (Univ. of Minn. L. Sch., Pre-Registration Statement, Sept. 21, 2025), <https://osf.io/h8s9d/overview>.

disappears once we control for the strength of the synthesis memo. Members of the AI-exposed and control groups tended to produce strong reasoning memos if they had previously produced a strong synthesis memo. AI helped a larger share of the AI-exposed group produce a strong synthesis memo compared to the control group. The AI-exposed group therefore outperformed the control group on the downstream reasoning task.

Our results provide important, though necessarily limited, evidence that the use of AI on legal tasks does not inevitably undermine law students' and junior lawyers' ability to engage in independent legal reasoning. However, our results represent only an initial step in understanding the impact of AI usage on legal reasoning. Our findings are subject to several limitations. Most notably, we relied on law students rather than practicing attorneys, and the experimental design required participants to complete artificially constrained legal tasks that differ from the broader, more open-ended assignments junior lawyers typically receive.²⁰ Moreover, it remains possible—and perhaps even likely—that AI use could undermine human understanding in legal contexts that differ meaningfully from those we examined here.²¹

Additional results from our experiment underscore the concern that AI use can undermine independent legal reasoning. During the revision task, all participants were instructed to use AI to revise and improve the memo they drafted during the application task. We designed the revision task to isolate the effects of AI when introduced after participants had already independently analyzed the problem. Contrary to our expectations, while AI use during the revision stage helped participants improve weak reasoning memos, it also led participants with strong reasoning memos to make those memos worse. We interpret this pattern as evidence that AI can, at least in some circumstances, displace or dilute legal reasoning, even among relatively strong performers.

Beyond the short-term risk that reliance on AI may weaken sound legal reasoning, our findings shed no light on the longer-term consequences of

²⁰ See Part III.E, *infra* (exploring limitations of our results). The long-term impact of AI use is especially important to understand, both because of its implications for legal education and because it may help explain why many law firms and lawyers continue to resist adopting these tools. See BEN W. HEINEMAN JR., *THE INSIDE COUNSEL REVOLUTION: RESOLVING THE PARTNER-GUARDIAN TENSION* (2016) (discussing how in-house lawyers serve both as business partners and as guardians of the enterprise).

²¹ See *id.*

sustained AI use for lawyers. Emerging evidence from other professional domains suggests that heavy dependence on AI can impair durable learning, particularly for individuals who have not yet independently developed core competencies.²² We therefore regard rigorous study of these longer-term effects as essential, especially as an increasing share of law students and junior lawyers rely on AI across a broad range of tasks.

The Article proceeds in four Parts. Part I surveys the existing literature on AI's actual and anticipated influence on legal practice, with particular attention to the concern that AI use may erode lawyers' substantive understanding of the law. Part II details our empirical design and methodology for testing how AI affects human legal reasoning. Part III reports the results of our experiment. Finally, Part IV considers the broader implications of these findings for legal practice and legal education. Drawing on our results, it offers three distinct guidelines for capturing AI's benefits while reducing the risk that its use undermines human reasoning: lawyers and law students should use AI only for tasks when they can independently assess, explain, and build on AI's output; they should confine AI use to narrow, well-defined components of a project; and they should avoid relying on AI when working under tight time constraints or cognitive fatigue.

I. Evidence on Lawyering and AI Usage

A rapidly expanding body of empirical research suggests that generative AI systems are materially reshaping a variety of professions. Randomized controlled trials and large-scale field studies in domains such as medicine, accounting, management consulting, and customer service consistently report meaningful improvements in both speed and output quality when professionals receive access to generative AI tools.²³ Improvements to productivity are often largest for routine analytical and drafting tasks, though some studies also find improvements in more complex forms of problem solving. Repeated predictions that generative AI progress would plateau have largely proven to be incorrect. Successive generations of frontier systems

²² See *supra* note 17 and accompanying text.

²³ See, e.g., Shakked Noy & Whitney Zhang, *Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence*, 381 SCI. 187, 190 (2023); Fabrizio Dell'Acqua et al., *Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality* 2 (Harvard Bus. Sch., Working Paper 24-013, 2023).

have demonstrated steady gains across a wide range of standardized benchmarks in recent years, including measures of reasoning, coding, and domain-specific knowledge.²⁴

This Part explains how these broader findings translate to the legal profession. Section A reviews empirical studies on the use of AI for lawyering tasks. Section B discusses the risks of using AI for lawyering tasks. Section C concludes by offering a theory of how AI use may impair independent legal reasoning.

A. Potential Productivity Gains from AI-Powered Lawyering

Like other professions, lawyering involves a range of tasks for which AI may be more or less well suited. Lawyers conduct legal research by identifying relevant legal principles in case law, statutes, treatises, and other sources. Then, they must summarize and synthesize the material to understand the governing doctrine and how it has been applied in prior cases. Finally, they apply those principles to the facts of a client's situation—evaluating the strength of potential claims or defenses and developing legal arguments on the client's behalf. In practice, these activities are highly iterative: lawyers typically move back and forth among research, synthesis, and application as their understanding of the law and the facts, and the interrelationship of the two, evolves.

Emerging empirical research demonstrates that AI is capable of improving junior lawyers' productivity on at least some of these tasks. The strongest evidence comes from randomized controlled trials comparing upper-level law students' performance on realistic lawyering tasks with and without AI assistance.²⁵ In an experiment conducted in late 2024, participants saw statistically significant productivity gains in completing five of six legal tasks as compared to participants who did not have access to AI.²⁶ The

²⁴ See, e.g., Tejal Patwardhan et al., GDPval: Evaluating AI Model Performance on Real-World Economically Valuable Tasks (Sept. 2025) (unpublished manuscript), <https://doi.org/10.48550/arXiv.2510.04374>.

²⁵ See Schwarcz et al., *supra* note 2. Randomized controlled trials of this sort are especially valuable in assessing AI's present implications for legal practice because they focus on the causal effect of AI access on human performance. This remains the key question for assessing the potential impact of AI on the legal profession, at least until AI systems become sufficiently advanced that humans trust them to autonomously and independently perform legal tasks.

²⁶ See *id.*

magnitude of these productivity gains ranged from roughly 50 to 130 percent, reflecting both faster completion times and higher-quality outputs. Another randomized controlled trial in mid-2023 found that participants with access to an earlier AI model completed lawyering tasks between 5 and 38 percent more quickly than those in the control group.²⁷ But unlike in studies testing more advanced AI models, access to AI did not have a statistically significant impact on the quality of participants' work product.²⁸

Several benchmarking studies—which evaluate the quality of AI produced responses without any human involvement—have also demonstrated AI's potential to positively impact lawyering. Modern AI models frequently generate work product that rivals, and in some instances exceeds, that produced by human lawyers.²⁹ For example, one recent assessment tested multiple commercial legal AI platforms on a range of realistic lawyering tasks, finding that the quality of AI generated output generally matched or exceeded the quality of work produced by a cohort of competent, mid-level attorneys.³⁰

Unfortunately, rigorous empirical evidence regarding the impact on practicing attorneys of adopting AI technologies is limited. One important explanation for this lack of data is that many lawyers and law firms are reluctant to disclose sufficiently granular data to make such studies feasible.³¹ Nevertheless, there is some evidence suggesting that the potential productivity benefits of using AI for lawyering translate to the real world. For instance, one study evaluated the impact of supplying approximately one hundred legal aid lawyers with a late 2024 AI model. A substantial majority of participating lawyers reported perceived improvements in productivity,

²⁷ Choi et al., *supra* note 2.

²⁸ *Id.* at 171.

²⁹ See Neel Guha et al., *Legalbench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models*, NEURIPS (2023), <https://arxiv.org/pdf/2308.11462>; Yu Fan et al., *LEXam: Benchmarking Legal Reasoning on 340 Law Exams* (Jan. 20, 2026) (unpublished manuscript), <https://arxiv.org/abs/2505.12864>; ERIC A. POSNER & SARAN SHIVAM, UNIV. OF CHI. L. SCH., COASE-SANDOR INST. FOR L. & ECON. RSCH. PAPER NO. 25-03, *JUDGE AI: ASSESSING LARGE LANGUAGE MODELS IN JUDICIAL DECISION-MAKING* (2025), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5098708.

³⁰ See *Vals Legal AI Report*, VALS.AI, <https://www.vals.ai/industry-reports/vlair-2-27-25> (last visited Mar. 21, 2026).

³¹ One financially feasible way to test this would be to use contract attorneys. See Part IV, *infra*.

and many indicated an intention to continue using the technology.³²

The potential productivity gains from using AI for certain lawyering tasks are likely to grow in the coming years. Major legal-research providers, such as Westlaw and LexisNexis,³³ are investing heavily in AI technologies, while numerous legal AI startups are entering the field. At the same time, the underlying AI foundation models continue to improve. Taken together, these developments suggest that the legal profession will increasingly confront practical questions about how best to integrate AI into legal work, including which tasks are most amenable to AI assistance and whether reliance on AI poses risks to short-term work quality or long-term skill development.

B. Potential Risks Associated with AI-Powered Lawyering

Despite the mounting evidence that AI usage can dramatically improve the productivity of law students and junior lawyers across a wide range of tasks, adoption within much of the profession remains uneven and cautious. This hesitancy undoubtedly stems from a mix of social, economic, and regulatory dynamics, many of which are not unique to the legal profession. Even so, when members of the profession articulate their concerns, they tend to converge on three central risks that, in their view, justify a measured approach to integrating AI into legal practice: the risk of hallucinations, confidentiality problems, and undermining human legal reasoning.

The most frequently invoked concern about integrating AI into legal practice is the risk that these systems will hallucinate legal authorities.³⁴ Although this risk continues to manifest itself in legal filings and even

³² See Chien & Kim, *supra* note 3.

³³ See Press Release, Thompson Reuters, Thomson Reuters Unveils Generative AI Strategy Designed to Transform the Future of Professionals (Nov. 1, 2023) (indicating intent to invest \$100 million per year in generative AI); *Lexis+ AI Fuels \$30M Revenue Growth in Law Firms, New Study Finds*, LEXISNEXIS (May 13, 2025), <https://www.lexisnexis.com/community/pressroom/b/news/posts/lexis-ai-fuels-30m-revenue-growth-in-law-firms-new-study-finds>.

³⁴ See Dahl et al., *supra* note 9, at 66; Magesh et al., *supra* note 9. At its most obvious, this problem involves citations to cases and other legal source materials that do not exist at all. More subtly, it can also include references to genuine decisions that do not support the propositions for which they are cited, or attribution of fabricated quotations to authentic sources. Instances of such fabricated authorities appearing in court filings have been widely covered in both general and specialized legal media. See, e.g., Benjamin Weiser, *Here's What Happens when Your Lawyer Uses ChatGPT*, N.Y. TIMES (May 27, 2023), <https://www.nytimes.com/2023/05/27/nyregion/avianca-airline-lawsuit-chatgpt.html>.

judicial opinions,³⁵ it can be mitigated through long standing and routine lawyering practices, such as verifying the existence, accuracy, and relevance of cited authorities in final versions of submitted materials.³⁶ Additionally, legal AI providers are aware of this concern and have invested in models specifically trained to facilitate and even automate cite-checking.³⁷ Nonetheless, AI may exacerbate the risk that attorneys will misread cases, overstate holdings, or cite authorities that do not squarely support their claims—particularly when using AI to generate legal documents quickly.

A second commonly cited concern is that lawyers' use of AI tools may compromise client confidences, constituting a breach of the lawyer's duties of confidentiality and secrecy, and even jeopardizing protections such as the attorney–client privilege or the work product doctrine.³⁸ As with the risk of hallucinated authorities, AI companies have invested heavily in technical safeguards designed to address these confidentiality concerns.³⁹ Many enterprise AI providers now offer contractual assurances that user data will be retained only for limited periods for safety monitoring and will not be used to train future models. Legal AI providers often go further, offering data

³⁵ See, e.g., *Hessert v. Hessert*, __ So. 3d __, __ (Fla. Dist. Ct. App. 2026) (“Courts across the United States, including Florida’s appellate courts, are currently grappling with an influx of court filings produced by pro se litigants and attorneys alike with the assistance of AI that cite non-existent cases or that cite actual cases for inaccurate legal propositions.”); Josh Blackman, *Two Federal Judges Apologize for Issuing Opinions with AI Hallucinations*, VOLOKH CONSPIRACY (Oct. 24, 2025), <https://reason.com/volokh/2025/10/24/two-federal-judges-apologize-for-issuing-opinions-with-ai-hallucinations/>.

³⁶ See Pa. Bar Ass’n Comm. on Legal Ethics & Pro. Resp. & Pa. Bar Ass’n Pro. Guidance Comm., *Joint Formal Op. 2024-200: Ethical Issues Regarding the Use of Artificial Intelligence* (May 22, 2024), <https://www.lawnext.com/wp-content/uploads/2024/06/Joint-Formal-Opinion-2024-200.pdf> (observing that some lawyers use generative AI “without checking citations and legal arguments” and thereby “fail to perform due diligence”); D.C. Bar Legal Ethics Comm., *Ethics Op. 388: Attorneys’ Use of Generative Artificial Intelligence in Client Matters* (Apr. 11, 2024), <https://www.dcbar.org/for-lawyers/legal-ethics/ethics-opinions-210-present/ethics-opinion-388> (warning that lawyers who “blindly rely on outputs produced by [generative AI] do so at considerable peril” and must “review and validate” outputs).

³⁷ See CITECHECK AI, <https://citecheck.ai> (last visited Mar. 21, 2026).

³⁸ See Andrew Perlman, *The Legal Ethics of Generative AI*, 57 SUFFOLK U. L. REV. 345, 347 (2024). Data retained by AI providers may also be subject to compelled disclosure through subpoena or other legal process, and, like any centralized data repository, may present an attractive target for unauthorized access in the event of a cybersecurity breach. Jonah E. Perlin, *Client Confidentiality and Generative AI*, 40 HARV. J.L. & TECH. (forthcoming 2027).

³⁹ See Perlin, *supra* note 38.

encryption that prevents any outside organizations from accessing user-supplied materials or prompts. Such safeguards are comparable to those that lawyers rely on when using cloud computing or email.⁴⁰

In contrast to concerns about hallucinations or confidentiality, the risk that reliance on AI may erode lawyers' underlying professional competence is far more difficult to manage through well-established practices and procedures.⁴¹ Lawyers have long understood that deep understanding of legal issues emerges from sustained engagement with primary sources: reading legal authorities closely, drafting and redrafting analyses, testing arguments through hypotheticals, and refining positions through pointed questioning and debate.⁴² These activities are not merely instrumental steps toward an answer; they are the means by which lawyers internalize doctrine, develop judgment, and cultivate strategic insight.⁴³

When human lawyers rely on AI to help perform or structure reasoning tasks, they risk short-circuiting a fundamental feature of the lawyering process.⁴⁴ This can have negative results in the short-term and in the long-term. In the short-term, lawyers who rely too heavily on AI may struggle to respond thoughtfully to probing questions from clients, judges, or

⁴⁰ *Id.* In matters involving particularly sensitive information, additional protections are available, including hosting AI systems on segregated or dedicated servers that isolate firm data from broader systems. When lawyers lack access to enterprise-grade protections, confidentiality risks can still be substantially mitigated through straightforward precautions, such as redacting or anonymizing sensitive facts, avoiding unnecessary identifying details, and carefully reviewing general AI suppliers' data governance terms.

⁴¹ See Am. Bar Ass'n Standing Comm. on Ethics & Pro. Resp., Formal Op. 512 (July 29, 2024), https://www.americanbar.org/content/dam/aba/administrative/professional_responsibility/ethics-opinions/aba-formal-opinion-512.pdf; N.Y.C. Bar Ass'n Comm. on Pro. Ethics, Formal Op. 2024-5: Ethical Obligations of Lawyers and Law Firms Relating to the Use of Generative Artificial Intelligence in the Practice of Law (2024), https://www.nycbar.org/wp-content/uploads/2024/08/20221329_GenerativeAILawPractice.pdf.

⁴² *Legal Research Strategy*, HARV. L. SCH. LIBR., <https://guides.library.harvard.edu/law/researchstrategy> (last visited Mar. 21, 2026).

⁴³ See Schwarcz et al., *supra* note 7; Debarati Das et al., LawFlow: Collecting and Simulating Lawyers' Thought Processes on Business Formation Case Studies (Apr. 26, 2025) (unpublished manuscript), <https://arxiv.org/abs/2504.18942>. See also Editorial, *Writing Is Thinking*, 3 NAT. REV. BIOENGINEERING 431 (2025), <https://doi.org/10.1038/s44222-025-00323-4>.

⁴⁴ See N.Y. STATE BAR ASS'N TASK FORCE ON A.I., REPORT AND RECOMMENDATIONS TO NYSBA HOUSE OF DELEGATES (Apr. 6, 2024), <https://nysba.org/wp-content/uploads/2022/03/2024-April-Report-and-Recommendations-of-the-Task-Force-on-Artificial-Intelligence.pdf>.

supervisors. They may overlook unstated assumptions, fail to anticipate counterarguments, or miss connections among related issues within a broader litigation or transactional strategy. Over the long term, such gaps could translate into diminished professional capacity in settings where AI is unavailable, constrained, or simply less helpful.⁴⁵

Although there is limited empirical evidence regarding these potentially negative consequences of AI use in the legal setting, there is mounting evidence demonstrating that it exists in other settings. For instance, one prominent MIT Media Lab experiment measured brain activity of young adults writing SAT-style essays using either ChatGPT, Google search, or no online tools. The group using ChatGPT showed the lowest brain activity and consistently underperformed on neural and behavioral metrics. As the study progressed, many ChatGPT users became increasingly passive – often copy-pasting AI-generated text into their final essays.⁴⁶ Participants who had relied on ChatGPT struggled to recall or explain the essay content afterward. When asked to rewrite their essays without AI, the ChatGPT group remembered very little of what they had “written,” indicating they had not internalized the material.

Another recent study finds similar experimental evidence that reliance on AI can undermine internal reasoning processes.⁴⁷ The study evaluated participants’ performance on reasoning problems designed to measure a person’s ability to override an immediate, intuitive, but incorrect answer and instead engage in more deliberate, analytical thinking. It found that participants frequently consulted an AI assistant and adopted its answers with minimal scrutiny. When the AI was experimentally manipulated to be accurate, participants’ performance improved substantially; when it was manipulated to be confidently wrong, accuracy dropped well below baseline levels, indicating that users often allowed AI outputs to displace human reasoning. Even more striking, participants followed faulty AI recommendations in the majority of cases and became more confident when using AI, including after incorrect responses.

⁴⁵ See Anna C. Conley, *Legal Education’s Role in Combating Automation Bias and Complacency with Generative AI in the Practice of Law*, KILAW J. (2024).

⁴⁶ Kosmyna et al., *supra* note 17.

⁴⁷ Steven D. Shaw & Gideon Nave, *Thinking Fast, Slow, and Artificial: How AI Is Reshaping Human Reasoning and the Rise of Cognitive Surrender* (Jan. 11, 2026) (unpublished manuscript), <https://ssrn.com/abstract=6097646>.

These findings that AI use can undermine human understanding are not isolated. Similar findings have emerged in a wide range of fields.⁴⁸ For instance, one experiment found that undergraduates who relied on an AI software coding tool were able to produce functional code more quickly than those who did not, but reported that they did not fully understand the code they produced.⁴⁹ Another study found that physicians who relied on AI-assisted diagnostic tools sometimes became overly dependent on them, deferring to incorrect AI assessments in ways that ultimately reduced their diagnostic accuracy when the tool was unavailable.⁵⁰ Similar results have been observed in the context of professional recruiters. Recruiters who used lower-quality AI systems were more accurate in evaluating resumes than recruiters who used higher-quality AI because they spent more time reviewing applications and were less likely to automatically follow the algorithm's recommendations.⁵¹

⁴⁸ See Hao-Ping (Hank) Lee et al., *The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects from a Survey of Knowledge Workers*, in *CHI Conference on Human Factors in Computing Systems 1* (2025).

⁴⁹ See Istiak Hossain Shihab et al., *The Effects of GitHub Copilot on Computing Students' Programming Effectiveness, Efficiency, and Processes in Brownfield Programming Tasks* (2025) (unpublished manuscript), <https://doi.org/10.48550/arXiv.2506.10051>. Preliminary descriptions of a follow-up study examining brain activity in programmers with and without AI assistance concluded that reliance on AI in coding may significantly reduce developers' active problem-solving, creativity, and critical thinking during programming. See Andrew R. Chow, *ChatGPT May Be Eroding Critical Thinking Skills, According to a New MIT Study*, TIME (June 23, 2025), <https://time.com/7295195/ai-chatgpt-google-learning-school> ("Kosmyna says that she and her colleagues are now working on another similar paper testing brain activity in software engineering and programming with or without AI, and says that so far, 'the results are even worse.'").

⁵⁰ See Drew Prinster et al., *Care to Explain? AI Explanation Types Differentially Impact Chest Radiograph Diagnostic Performance and Physician Trust in AI*, 313 RADIOLOGY 2 (2024), <https://doi.org/10.1148/radiol.233261>. See also M. Prochaska & David Alfandre, *Artificial Intelligence, Ethics, and Hospital Medicine: Addressing Challenges to Ethical Norms and Patient-Centered Care*, 19 J. HOSP. MED. 1194 (2024); Krysztuf Budzyń et al., *Endoscopist Deskillng Risk After Exposure to Artificial Intelligence in Colonoscopy: A Multicentre, Observational Study*, 10 LANCET GASTROENTEROLOGY & HEPATOLOGY 896 (2025), [https://doi.org/10.1016/S2468-1253\(25\)00133-5](https://doi.org/10.1016/S2468-1253(25)00133-5); F. Yu, A. Moehring, O. Banerjee et al., *Heterogeneity and Predictors of the Effects of AI Assistance on Radiologists*, 30 NATURE MED. 837 (2024).

⁵¹ See Fabrizio Dell'Acqua, *Falling Asleep at the Wheel: Human/AI Collaboration in a Field Experiment on HR Recruiters* (2022) (unpublished manuscript), https://www.dropbox.com/scl/fi/c5jdi146erky69csa8gha/Falling_Asleep_at_the_Wheel.pdf?rlkey=ss4vh6bsgzf52xiba8ffx0onc&st=s0ppis9g&dl=0.

C. *Theoretical Expectations*

Using the existing empirical literature and the concerns raised by the legal profession, we develop a theory to explain how AI use for some tasks, such as summarizing legal texts, may have downstream consequences on independent legal reasoning. By reducing the level of cognitive engagement with the underlying sources, we predict that lawyers experience a decline in their own independent ability to engage in reasoning tasks. This results in a tradeoff: AI use increases productivity on tasks while impairing performance on subsequent tasks when AI is no longer available. Based on this initial theory, we developed preregistered hypotheses, which we further explain in Part II.

Existing empirical research indicates that the use of AI can improve both the quality and efficiency with which junior lawyers perform many common legal tasks. Summarizing and synthesizing large corpora of text is one of AI's core competencies.⁵²

Synthesis Task Quality Hypothesis: Lawyers who use AI to synthesize legal texts will produce higher-quality summaries than lawyers who do not use AI.

Synthesis Task Speed Hypothesis: Lawyers who use AI to synthesize legal texts will produce summaries in less time than lawyers who do not use AI.

By relying on AI for initial tasks, lawyers are less cognitively engaged with the underlying legal texts. Improvements in quality and speed associated with AI come at the downstream cost of lawyers' ability to perform legal reasoning tasks that rely on a deep understanding of those texts when they no longer have access to AI. The erosion of legal-reasoning skills may occur through two pathways: recall or comprehension.

Legal reasoning requires lawyers to recall and comprehend nuanced information about legal principles. This skill is on display, for example, when a lawyer responds to a judge's question about a particular case during oral argument. Consistent with emerging research suggesting that AI use can

⁵² Importantly, we do not assume that AI can produce high-quality legal research without oversight. Our framework focuses exclusively on AI use within a human-in-the-loop framework, in which lawyers use AI as a tool.

diminish users' understanding of underlying material,⁵³ lawyers who rely on AI to synthesize legal texts may struggle to recall key details from those texts or identify nuances that emerge from a careful synthesis.

Comprehension Task Hypothesis: Lawyers who use AI to synthesize legal texts will be less able to comprehend the legal principles contained in those texts than lawyers who do not use AI.

Comprehension alone, however, is insufficient for effective legal reasoning. The lawyer's core task is not merely to understand black-letter law, but to apply that law to novel factual scenarios. By outsourcing the initial analysis to AI, lawyers may be less capable of applying those rules to new or complex fact patterns, limiting their ability to adapt principles to novel situations. As a result, downstream work product for which AI is not relied upon, such as legal briefs or client memos, may show diminished quality in terms of analytical depth.

Application Task Hypothesis: Lawyers who use AI to synthesize legal texts will be less able to apply legal principles to novel factual scenarios than lawyers who do not use AI.

Additionally, the lack of initial engagement with the texts may impair the legal reasoning process by limiting the internalization of legal principles. Lawyers who have not internalized legal principles may need to refer more frequently to the source material while preparing client memos, legal briefs, or other documents that require legal reasoning. Accordingly, the speed gains from AI on these initial tasks may be offset by reduced efficiency in subsequent legal reasoning.

Application Task Speed Hypothesis: Lawyers who use AI to synthesize legal texts will take longer to apply legal principles to novel factual scenarios than lawyers who do not use AI.

Our theory focuses on the erosion of a lawyer's ability to engage in independent legal reasoning without the use of AI. This possibility is

⁵³ See, e.g., Judy Hanwen Shen & Alex Tamkin, How AI Impacts Skill Formation (Jan. 28, 2026) (unpublished manuscript), <https://arxiv.org/abs/2601.20245> ("We find that AI use impairs conceptual understanding, code reading, and debugging abilities, without delivering significant efficiency gains on average. Participants who fully delegated coding tasks showed some productivity improvements, but at the cost of learning.")

especially relevant when lawyers or law firms limit AI use at later lawyering stages or when the context limits practical access to any technology—for example, during oral arguments, client interviews, trials, and depositions. Nevertheless, in many situations, attorneys who rely on AI for initial tasks may also be expected to use AI when performing subsequent related tasks.

Given this reality, the final question is whether permitting AI use for later lawyering tasks could improve the quality of work product, potentially offsetting some of the losses caused by any decline in independent legal reasoning. While we generally expect that using AI for revisions will improve the quality of work product, the effects should be greatest among lawyers whose work product has the most room for improvement. These lawyers may benefit more from AI's ability to identify missing issues, clarify analytical structure, and suggest alternative applications of legal principles. By contrast, lawyers who already produce high-quality work product may experience smaller gains because AI primarily refines rather than fundamentally transforms high-quality legal analysis.

Revision Task Hypothesis: Lawyers who use AI to revise existing work product will experience improvements in the quality of their legal work, with the greatest benefits observed among those who initially produced weaker work product.

II. Experimental Design

We tested our theory using a randomized controlled trial. Randomized controlled trials are the gold standard for identifying the causal effects of an intervention—such as access to AI—because random assignment minimizes selection bias and balances observable and unobservable characteristics across groups.⁵⁴ In contrast, observational studies of legal practice are vulnerable to substantial selection effects: lawyers who choose to use AI on lawyering tasks may differ systematically from those who do not in ways that affect performance. Moreover, real-world outcomes—such as case success, client satisfaction, or billable hours—are shaped by numerous factors beyond the quality of legal reasoning, making it difficult to isolate the independent

⁵⁴ See JOSHUA D. ANGRIST & JÖRN-STEFFEN PISCHKE, *MOSTLY HARMLESS ECONOMETRICS: AN EMPIRICIST'S COMPANION* 15 (2009). For a greater discussion of our identification strategy, see *infra* App. A.1.

effect of AI use.

Consistent with the theory, our study assessed performance on four legal tasks performed by upper level law students: a synthesis task, comprehension task, application task, and revision task. Each task related to a problem involving servitudes that burden personal property. Prior to the start of the experiment, we randomly assigned participants to a treatment group that had access to AI for the synthesis task (i.e., the AI-exposed group) or a control group that did not.⁵⁵ Participants first read a source packet and wrote a memo summarizing the relevant doctrine. After writing their synthesis memo, participants completed a closed-book multiple choice quiz that tested their comprehension of the source packet. Participants then wrote a second memo, without AI, applying their knowledge of servitudes to a set of provided facts. Finally, participants used AI to help revise their second memo. Table 1 describes the four tasks and provides the operationalized versions of our theoretical hypotheses.

TABLE 1: TASK SUMMARY

Task	Format	Access to AI	Empirical Hypotheses
Synthesis	Memo summarizing source packet.	AI-exposed group but not control group.	<i>Synthesis Task Quality Hypothesis:</i> Participants in the AI-exposed group will receive higher scores on the synthesis task than participants in the control group. <i>Synthesis Task Speed Hypothesis:</i> Participants in the AI-exposed group will complete the synthesis task in less time than

⁵⁵ For a discussion of the composition of our group, see *infra* App. A.3.

			participants in the control group.
Comprehension	Multiple choice quiz.	Neither AI-exposed group nor control group.	<i>Comprehension Task Hypothesis:</i> Participants in the AI-exposed group will receive lower scores on the comprehension quiz.
Application	Memo applying knowledge from the source packet to a set of facts.	Neither AI-exposed group nor control group.	<i>Application task Quality Hypothesis:</i> Participants in the AI-exposed group will receive lower scores on the application task than participants in the control group. <i>Application task Speed Hypothesis:</i> Participants in the AI-exposed group will complete the application task in more time than participants in the control group.
Revision	Revised memo from the prior task.	Both AI-exposed group and control group	<i>Revision Hypothesis:</i> All participants will experience improvements in the quality of their reasoning memos, with the greatest benefits observed among those who produce weaker reasoning memos.

Section A discusses how we designed the study's general parameters, including selection of the skills to assess, framing of tasks, and development and preregistration of hypotheses. Section B provides details about specific design choices, such as topic selection, participant recruitment, time limits, and grading.

A. General Study Design

1. Selecting Skills to Assess

Our goal was to simulate a legal assignment for which AI is commonly employed. We focused on the kinds of assignments that senior lawyers often delegate to junior colleagues. Studying the effects of AI on junior lawyers was prudent given that we were limited to using law students as participants. These participants are closer in age and experience to junior lawyers than senior lawyers.

We selected an appropriate assignment by consulting a recent study that surveyed 15,000 lawyers to identify tasks that “newly licensed lawyers” (NLLs) typically perform.⁵⁶ The study fragmented these tasks into their component skills.⁵⁷ The study of NLLs identified four relevant skills that were amenable to testing in a standardized experiment: “read and understand information presented in writing,” “spot salient legal concerns presented by a set of circumstances,” “use analytical skills, logic, and reasoning to solve problems and to formulate advice,” and “effectively communicate information and ideas in writing.”⁵⁸

2. Identifying Four Tasks for Participants to Complete

We designed the experiment to do more than merely test whether AI enhanced or degraded the quality of a final work-product. We also wanted to test whether AI had downstream consequences on legal reasoning even after lawyers stopped using AI. To do so, we designed an experiment with four distinct components: a synthesis task, comprehension task, application task, and revision task.

a. Synthesis Task

⁵⁶ NATIONAL CONFERENCE OF BAR EXAMINERS, FINAL REPORT OF THE TESTING TASK FORCE 2 (Apr. 2021). The NCBE defines NLLs as “lawyers who have been licensed for three years or less.” *Id.* at 6.

⁵⁷ *See id.* at 9–11.

⁵⁸ *Id.* at 11.

The synthesis task was designed to test whether using AI can help lawyers synthesize legal sources addressing unfamiliar subjects. We cast each participant in the role of a law firm associate who received an email from a partner asking them to summarize a legal rule based exclusively on five supplied sources.⁵⁹ The partner explained that: “Your memo should outline the elements of the rule and any exceptions, providing a framework for how a court would approach the legal question. In other words, don’t merely summarize the sources; synthesize them into a summary of the rule that indicates how the elements fit together.”⁶⁰ Participants had up to 75 minutes to read the packet and complete the memo.⁶¹ The control group was instructed not to use AI for this task, while the AI-exposed group was instructed to use Gemini 2.5 Pro⁶² “to assist you in writing the assignment.”⁶³

b. Comprehension Task

We anticipated that during the synthesis task, the control group would closely read the source packet to identify all relevant nuances. In contrast, we anticipated that the treatment group might read the source packet less carefully. They might rely on AI as a crutch, or they might become distracted by the need to engineer prompts and evaluate AI output. To measure these potential effects, we designed a comprehension task consisting of six multiple choice questions that participants would answer in ten minutes without access to either the source packet or AI.⁶⁴ We designed the questions to be

⁵⁹ For a description of the sources, see *infra* Part II.B. The sources were contained in a twelve-page (4457-word) “Source Packet.” The instructions required participants to rely exclusively on the provided source packet.

⁶⁰ For the full instructions for the synthesis task, see *infra* App B.1.

⁶¹ There was no word limit, but the partner suggested writing approximately 750 words. This limit was necessary to ensure that AI did not enable participants in the AI-exposed group to write longer memos than participants in the control group.

⁶² We selected Gemini 2.5 Pro as the required AI for two reasons. First, at the time of the study it was state-of-the-art. See Guha et al., *supra* note 29. For current LegalBench data, see *LegalBench*, VALS.AI, https://www.vals.ai/benchmarks/legal_bench (Mar. 18, 2026). Second, it was accessible to participants without cost through the University of Minnesota.

⁶³ *Id.* Participants could incorporate AI-generated text into their memos and could upload their drafts to AI and ask for suggestions and revisions. The instructions gave specific guidance about how to craft a prompt that would confine AI’s suggestions to the closed universe of the source packet. Third, the university’s subscription facilitated creation of a closed-universe test that ensured data privacy.

⁶⁴ For the full list of questions, see *infra* App. C.1.

moderately challenging if a participant carefully read the entire packet, but to be more difficult if the participant read the packet less carefully.

c. Application Task

We wanted to assess how the use of AI on the synthesis task affected participants' ability to apply the results of the synthesis task to a reasoning task. Accordingly, both the control and treatment groups performed the same application task without AI. That task presented participants with a follow-up email from the partner who had assigned them the synthesis task instructing them to write a memo applying their knowledge from the synthesis task to a new set of facts.⁶⁵ Participants had access to their prior synthesis memo and were instructed to save time by not repeating its contents. The new reasoning memo should instead: "identify strengths and weaknesses in the client's position, recommend arguments that the client should make, and rebut counterarguments."⁶⁶ Participants had up to 60 minutes to complete the memo.⁶⁷

d. Revision Task

We were curious whether using AI to revise a previously completed memo would enhance or diminish the quality of the final product. We therefore designed a revision task, in the form of another email from the supervising partner, which asked participants to revise their memo from the application task so that it is "clearer and better written."⁶⁸ The partner instructed the associate: "Do not alter the substance of the memo, though you should feel free to elaborate on or better explain any points in the memo."⁶⁹ All participants were instructed to use AI to assist in this task. Participants received guidance about crafting prompts that would preserve the closed universe of the experiment. The instructions stated that participants could "accept all, some, or none of Gemini's comments or suggestions" and should

⁶⁵ For the full instructions for the application task, see *infra* App. D.1.

⁶⁶ *Id.*

⁶⁷ There was no recommended word limit to avoid placing an artificial constraint on participants' analysis.

⁶⁸ For the full instructions for the revision task, see *infra* App. E.1.

⁶⁹ *Id.*

“make sure that any incorporation of Gemini’s suggestions faithfully reflects your intended meaning.”⁷⁰ Participants had 20 minutes to complete the revision task.

B. Specific Design Choices

After identifying our research goals and designing the study, we resolved several subsidiary issues. Section 1 explains how we designed the memo topic and source packet. Section 2 documents the process of recruiting and compensating participants. Section 3 discusses how we structured time limits. Section 4 outlines the process of creating and applying the grading rubric.

1. Topic Selection and Source Packet

Four criteria guided our selection of substantive materials. First, participants should not have prior familiarity with the topic, so as to avoid the risk that subject matter knowledge would be unevenly distributed among participants. Second, participants should understand the context in which the topic arises to better reflect the reality of legal practice and ensure that the study’s tight time limits did not foreclose meaningful analysis. Third, the topic needed to be sufficiently complex to challenge analytical skills, yet not so opaque that it would trigger widespread participant confusion. Fourth, the topic should require participants to assess multiple factors so as to provide several opportunities for AI to influence discrete aspects of a participant’s analysis.

We relied on these four criteria to construct a hypothetical problem about servitudes that burden personal property. Servitudes on real property are rights and obligations that “run[] with” the affected estate.⁷¹ They therefore bind both the owner who created the servitude and subsequent purchasers.⁷² Easements and covenants are prominent examples of servitudes.⁷³ In theory, servitudes can also burden personal property. For example, a contract

⁷⁰ *Id.*

⁷¹ RESTATEMENT (THIRD) OF PROPERTY: SERVITUDES § 1.1 cmt. a (AM. L. INST. 2000).

⁷² *See id.*

⁷³ For example, if Owner A grants Neighbor B a recorded easement allowing B to use a driveway that crosses A’s land to access a public road, that easement “runs with” the property. When A later sells the land to Purchaser C, C takes title subject to the easement and must continue to allow B to use the driveway. *See id.* at § 1.1(2).

conveying ownership of a chattel might attempt to forbid a particular use or create a right of access for someone other than the owner.

We concluded that this topic met our four initial criteria. While obscure,⁷⁴ the topic is conceptually adjacent to the 1L Property curriculum at the University of Minnesota Law School.⁷⁵ Moreover, the doctrine is reasonably straightforward, but requires analyzing several variables, including the clarity of the text purporting to create a servitude, the scope of the servitude, the purpose of the servitude, whether a subsequent purchaser had adequate notice of the servitude, and whether the servitude unduly restricts alienation of the property or otherwise undermines public interests.

After selecting the topic of servitudes that burden personal property, we constructed a source packet and hypothetical scenario. The source packet was twelve pages long. It included five edited sections of the *Restatement (Third) Property* that defined servitudes, and four judicial opinions from a hypothetical fifty-first state of the United States.⁷⁶ We wrote each opinion, drawing elements from both scholarship⁷⁷ and real cases.⁷⁸ Collectively, the opinions included partial doctrinal tests, exhibited a shift in doctrinal

⁷⁴ The application of servitudes to personal property is sufficiently obscure that the *Restatement (Third) of Property* declined to address the topic. *See id.* at § 1.1 cmt. b (“Whether servitudes can be created to burden or benefit personal property has been the subject of some debate, but little litigation. No position is taken on the question in this Restatement.”). Similarly, a small body of scholarship discusses servitudes on personal property. *See, e.g.,* Danielle D’Onfro, *Contract-Wrapped Property*, 137 HARV. L. REV. 1058 (2024); Christina Mulligan, *Personal Property Servitudes on the Internet of Things*, 50 GA. L. REV. 1121 (2016); Glen O. Robinson, *Personal Property Servitudes*, 71 U. CHI. L. REV. 1449 (2004); Zechariah Chafee, Jr., *The Music Goes Round and Round: Equitable Servitudes and Chattels*, 69 HARV. L. REV. 1250 (1956); Zechariah Chafee, Jr., *Equitable Servitudes on Chattels*, 41 HARV. L. REV. 945 (1928). Given our knowledge of the property-law curriculum at the University of Minnesota, our study participants were unlikely to have read this literature or had prior exposure to servitudes burdening personal property.

⁷⁵ Law students in our study cohort had prior exposure to servitudes on real property, but did not have prior exposure to servitudes on personal property. We reviewed the syllabi for 1L Property courses at the University of Minnesota Law School—from which we drew all our participants—during the 2023–24 and 2024–25 academic years. All the syllabi covered servitudes that attached to land; none covered servitudes that attached to personal property. The syllabi are on file with the authors.

⁷⁶ We relied on a nonexistent jurisdiction to mitigate the risk of participants and AI incorporating knowledge outside the experiment’s bounds.

⁷⁷ *See* sources cited *supra* note 74.

⁷⁸ *See* *Nadell & Co. v. Grasso*, 346 P.2d 505 (Cal. Dist. Ct. App. 1959); *Pratte v. Balatsos*, 113 A.2d 492 (N.H. 1955). Two of the opinions included a separate dissent or concurrence that added conflicting viewpoints for participants to analyze.

emphasis over time, identified and weighed competing policies, offered extensive dicta of debatable significance, and resulted in nominally inconsistent outcomes that participants would need to harmonize or explain. The opinions did not spotlight key takeaway points, and included both relevant and irrelevant information.

We also developed a hypothetical scenario that we revealed only in the Application Task. This scenario concerned the sale of a valuable antique car that had appeared in a famous film. The sales contract purported to prohibit subsequent purchasers from materially altering the car's appearance. Participants were required to assess the validity of this restraint on use of the chattel.

After designing these materials, we hired several research assistants to pilot them. This enabled us to test the source packet and hypothetical problem, and to refine our rubrics to evaluate performance across the four tasks.

2. *Recruitment of Participants*

We recruited 91 second- and third-year students from the University of Minnesota Law School.⁷⁹ Participants who completed the experiment received a fixed payment of \$75. To incentivize effort, participants scoring in the top forty percent of the treatment or control group (as applicable) received an additional \$50, and those scoring in the top ten percent received an additional \$100.⁸⁰ We collected basic information from participants, including their first-year GPA, their year in law school, and whether they had any prior exposure to AI. We randomly assigned participants to either a control group (N = 41) or a treatment group (N = 50).⁸¹ There were no statistically distinguishable differences between the control group and the

⁷⁹ We excluded first-year students and LLM students. We did so for two reasons. First, their ability to synthesize and apply relevant concepts is generally less uniform and less advanced than the ability of students who have had more time to study U.S. law. Second, the excluded groups would have had inconsistent prior exposure to property law and servitudes.

⁸⁰ This payment structure is consistent with best practices.

⁸¹ The treatment group was larger due to attrition within the control group. Attrition occurred prior to exposure to the assignments and therefore the higher rate of attrition in the control group appears to have been random and unrelated to any aspect of the experiment. Moreover, the attrition did not affect the balance between the two groups. *See infra* App. A.3.

treatment group with respect to the information we collected.⁸²

3. Time Limits

We limited the experiment to three hours, consistent with the typical length of law school exams, which also helped facilitate participant recruitment.⁸³ To approximate real-world practice conditions, we gave participants the option of spending less than the maximum time allowed for each task.⁸⁴

Participants completed the experiment remotely in October, 2025, by logging into a Canvas site that contained all required instructions and materials.⁸⁵ The control and treatment groups used separate Canvas sites, which were identical except for instructions related to the use of AI on the synthesis task. Participants completed the experiment in a single sitting and reported their time to complete each task.

⁸² See *infra* App. A.3.

⁸³ We initially considered trying to ensure uniformity by requiring every participant to spend the maximum allowed time on each task. Such a requirement, however, would have prevented us from observing whether AI produces efficiency gains. Moreover, there was no way to enforce this requirement, and any assumption of compliance would have been questionable.

⁸⁴ The instructions to participants explained: “Each assignment has a maximum time limit, but you can finish the assigned work before that time limit. You should approach each assignment as if you are a junior attorney who is producing work for a fee-sensitive client. While you can take up to the maximum time allotment to complete each task, you should upload your memo at the point where you would feel comfortable submitting your work product to a supervising attorney, given that your client would prefer to minimize the amount they pay for your work product. If you reach the end of the maximum time and have not finished, you should upload the work you were able to produce within the allotted time. Do not spend any more than the maximum time on any assignment.” See *infra* App. A.4 at ¶ 8.

⁸⁵ The instructions required participants not to discuss the experiment with anyone during the window when the experiment was live. All participants were instructed not to use AI for tasks (2) and (3), and participants in the control group were also instructed not to use AI for task (1). See Part II.A, *supra*. However, we were not able to technologically limit participants’ ability to access AI, meaning there is a risk that at least some participants used AI on tasks notwithstanding our instructions. This possibility has the greatest potential to compromise our results if participants in the AI-treatment group used AI on task 3 notwithstanding our instructions that they not do so. However, two features of our results suggest this was not a widespread problem. First, if the AI-assisted students had used AI on task (3), we would have expected them to complete the task faster than the control group. However, the AI-assisted students took longer to complete task (3) than the control group. See Part III, *infra*. Second, once we controlled for the participant’s initial score on task (1), the difference between the control and treatment groups is not predictive of performance on task (3). See *id*.

We received clearance from the University of Minnesota Institutional Review Board⁸⁶ and preregistered our hypotheses with the Center for Open Science prior to analyzing the data.⁸⁷

4. Grading

Accurate grading of participants' synthesis, application, and revision memos was essential to generating valid data. Pure objectivity is impossible when assessing written analysis of complex legal topics. But we took several steps to minimize subjectivity. First, we graded memos using preregistered rubrics,⁸⁸ which identified the essential substantive points that quality memos should address, the important nuances that weak memos might overlook, and the mistakes that a confused lawyer might make. Second, the rubrics included separate metrics for style, organization, and polish. Third, to ensure consistency and rigor, a single researcher, without access to any of the participant data, graded all 273 memos in November 2025. This researcher has more than twenty years of experience as a legal writing professor and law school faculty member. Finally, all memos were fully anonymized and randomized before grading, so that the grader did not know whether the participant was in the control or treatment group, or how the participant performed on any other tasks.

III. Results

This Part presents the empirical results of our study. We hypothesized that participants in the AI-exposed group would outperform those in the control group on the synthesis tasks but underperform on the comprehension and application tasks. The results provide partial support for these hypotheses. Consistent with prior research,⁸⁹ participants in the AI-exposed group received higher scores on the synthesis task and completed these tasks faster than participants in the control group.⁹⁰ Contrary to our expectations, however, participants in the AI-exposed group performed *better* than those in

⁸⁶ University of Minnesota IRB, IRB ID STUDY00025862, July 18, 2025.

⁸⁷ Nicholas Bednar et al., *Legal Reasoning and Artificial Intelligence*, OSF (Oct. 28, 2025), <https://osf.io/h8s9d/overview>.

⁸⁸ See *infra* note 96.

⁸⁹ See *infra* Part I.A.

⁹⁰ See *infra* Part III.A.

the control group on the application task even though neither group had access to AI for this task.⁹¹ These findings hold when we control for other possible explanations, such as GPA, year in law school, and prior AI usage.⁹²

What mechanism explains the stronger performance of participants in the AI-exposed group on the application task? We find that a participant's score on the synthesis task is highly predictive of their subsequent score on the application task.⁹³ In other words, the ability to produce a strong synthesis memo appears to have downstream benefits on legal reasoning. Because AI assistance improves performance on the synthesis task, it appears to indirectly enhance subsequent performance on the application task even when AI is no longer available.

We also hypothesized that all participants would see a boost in their performance when they were allowed to use AI to revise their work in the Revision task, with the lowest-scoring participants improving the most from using AI.⁹⁴ Somewhat surprisingly, however, participants who scored above the mean on the application task received lower scores after revising their memos with AI. Although our evidence here is limited, the result suggests that AI may hinder legal reasoning among participants who already demonstrate strong legal-reasoning skills, particularly under time constraints while suffering cognitive fatigue.

A. Results for the Synthesis Task

Prior research demonstrates that AI excels at summarizing and synthesizing legal texts.⁹⁵ Consequently, the synthesis task hypotheses predict that participants in the AI-exposed group will perform better on the synthesis task than those in the control group. For each task involving the production of a written memo, we evaluated performance in several important ways. First, we measured the *quality* of the work product by having a single researcher anonymously grade each memo using a standardized rubric.⁹⁶ Second, we measured the *speed* of the participant by examining the

⁹¹ See *infra* Part III.C.

⁹² See *infra* App. D.

⁹³ See *infra* Part III.C.

⁹⁴ See *infra* Part III.D.

⁹⁵ See Daniel Schwarcz & Jonathan H. Choi, *AI Tools for Lawyers: A Practical Guide*, 108 MINN. L. REV. HEADNOTES 1, 12–14 (2023).

⁹⁶ All four researchers participated in developing the standardized rubric. The use of a

time it took them to complete the task. Third and finally, we measured total *productivity* by dividing the participant's overall score by the number of minutes it took them to complete the task. This final measure allows us to evaluate trade-offs that may emerge between quality and speed when participants use AI to produce written work.⁹⁷

The *Synthesis Task Quality Hypothesis* predicts that participants in the AI-exposed group will produce higher quality synthesis memos than participants in the control group. During the first task, participants read a packet of source materials and synthesized the key legal principles in a short memo.⁹⁸ To measure quality, each memo was evaluated along three dimensions:

- *Substantive Quality* (Scored: 1-5): Overall impression of whether the memo provides an adequate explanation of the important substantive variables likely to affect the outcome of the case. (Sample Mean = 2.64, Sample SD = 1.18).⁹⁹
- *Organization* (Scored: 1-5): Overall impression of whether the memo organizes the material in an effective and reasonable matter. (Sample Mean = 3.15, Sample SD = 1.29).
- *Polish* (Scored: 1-5): Overall impression of the polish of the writing, such as complete headers, full citations, and finished prose. (Sample Mean = 3.02, Sample SD = 1.14).

On each dimension, a score of five reflects the highest possible performance, whereas a score of one reflects the lowest. We added these three components to produce a composite measure of overall quality (Sample Mean = 8.81, Sample SD = 3.30). To estimate the average treatment effect, we use a difference-in-means test to assess whether the memos produced by the AI-exposed group and the control group differed in quality.¹⁰⁰

single grader helped promote consistency and reduce measurement error that may arise with multiple graders.

⁹⁷ See Choi, Monahan & Schwarcz, *supra* note 2, at 153 (finding marginal and inconsistent improvements in quality but substantial improvements in efficiency).

⁹⁸ See *infra* Part II.A.1.

⁹⁹ Throughout, we assume the reader's familiarity with basic summary statistics, such as mean and standard deviation. For those who need a refresher on these concepts, see generally See JEFFREY M. WOOLDRIDGE, *INTRODUCTORY ECONOMETRICS: A MODERN APPROACH* (7th ed. 2016).

¹⁰⁰ See ANGRIST & PISCHKE, *supra* note 54, at 14–15 (explaining how to estimate the average treatment effect among two conditions in a randomized experiment). We estimated

TABLE 2: SYNTHESIS TASK: QUALITY

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	Effect (% Change)	P-Value
Overall	6.63 (2.42)	10.60 (2.84)	+3.97 (+59.78%)	<0.001***
Substantive Quality	1.95 (0.92)	3.22 (1.06)	+1.27 (+65.03%)	<0.001***
Organization	2.29 (0.96)	3.84 (1.09)	+1.55 (+67.49%)	<0.001***
Polish	2.39 (0.95)	3.54 (1.01)	+1.15 (+48.10%)	<0.001***

Note: P-value estimated using a two-tailed t-test.

Table 2 presents the effects of AI assistance on participants' performance on the synthesis task. Figure 1 depicts the distribution of the overall quality on the synthesis task for both groups. The results support the *Synthesis Task Hypothesis*. On average, participants in the control group scored 6.63 points, whereas those in the AI-exposed group scored 10.60 points. In other words, AI usage improved overall performance by 59.78%. Expressing the difference as a percentage helps illustrate the magnitude of the effect in intuitive terms. Because the raw scores are tied to the rubric used in this study, a percentage increase allows us to more easily understand the improvement experienced by participants who had access to AI. Participants who were assisted by AI in the production of their synthesis memos also exhibited improvements in substantive quality, organization, and polish.

The effect sizes are striking and meaningfully exceed those reported in prior studies testing earlier AI models. Standard deviations measure effect size by showing how large the difference between two groups is relative to the typical variation within those groups, which facilitates comparison of results across different studies.¹⁰¹ In training and education research, improvements of one-quarter to one-half of a standard deviation are typically considered meaningful.¹⁰² Prior research has found that access to AI

the statistical significance of the difference-in-means using a two-tailed t-test.

¹⁰¹ See WOOLDRIDGE, *supra* note 99, at 184 (explaining the value of measuring effect sizes in terms of standard deviations).

¹⁰² See Jon Veloski et al., *Systematic Review of the Literature on Assessment, Feedback and Physicians' Clinical Performance: BEME Guide No. 7*, 28 MED. TEACHER 117 (2006);

improves performance on lawyering tasks by about a third of a standard deviation.¹⁰³ By comparison, we observe an increase of more than one standard deviation (1.20 standard deviations) in overall quality on the synthesis task. Framed differently, the average participant in the control group scored in the 25th percentile. By contrast, the average participant in the AI-exposed group scored in the 71st percentile.

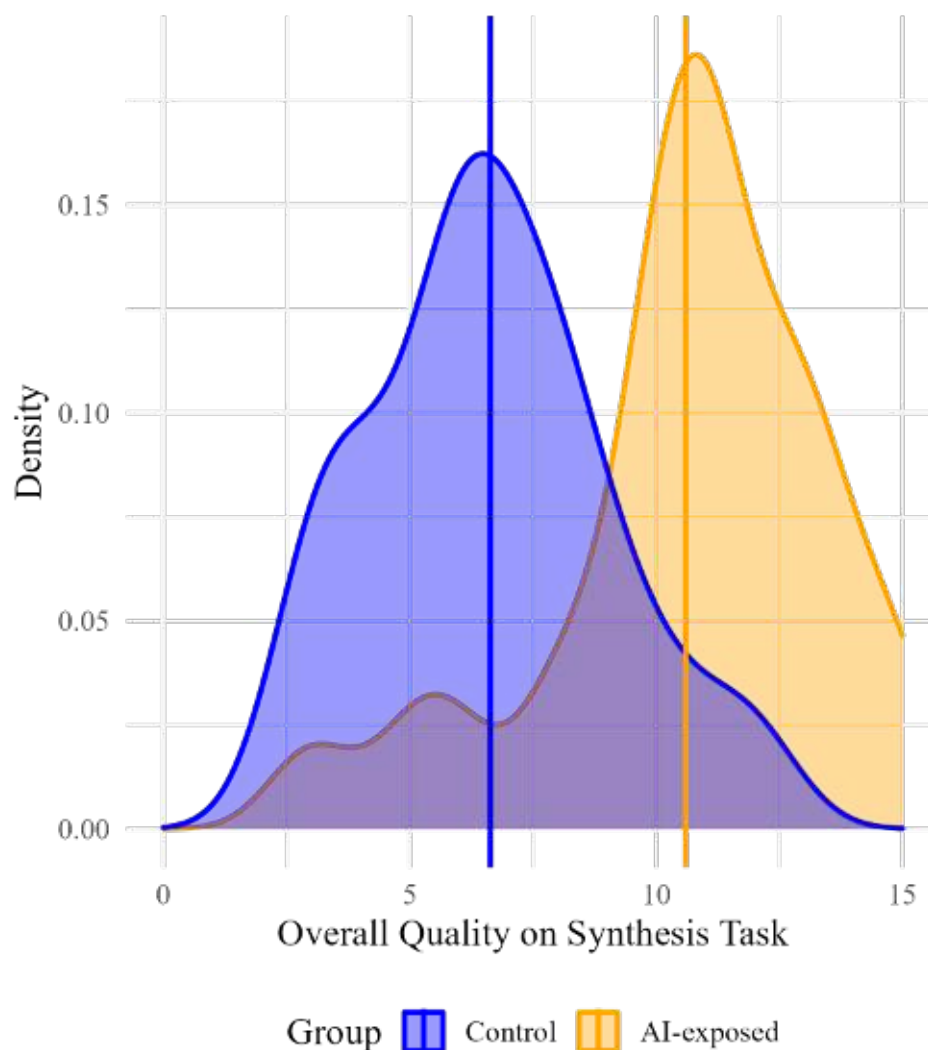
The observed improvements in overall quality are not driven by organization and polish alone. We also observe an increase of more than one standard deviation (1.08 standard deviations) in substantive quality. This indicates that AI is improving both form and substance. The larger gains observed here may reflect the rapid advancement of AI technologies, which have become increasingly capable of performing complex reasoning and writing tasks.¹⁰⁴ The effect may also reflect the possibility that AI assistance has a particularly powerful effect when the task requires synthesis, organization, and communication of legal material.

Catherine H. Crouch & Eric Mazur, *Peer Instruction: Ten Years of Experience and Results*, 69 AM. J. PHYSICS 970 (2001).

¹⁰³ See Schwarcz et al., *supra* note 2 (manuscript at 11).

¹⁰⁴ See Magesh et al., *supra* note 9, at 217 (showing the prevalence of hallucinations in AI tools designed for legal research).

FIGURE 1: OVERALL SCORE DISTRIBUTION ON SYNTHESIS TASK



Because participants were randomly assigned to each condition, the research design eliminates the selection bias that could arise if students were allowed to choose whether to use AI.¹⁰⁵ Nevertheless, we estimated ordinary least squares (OLS) regression models with controls for the student's GPA, year in law school, and prior AI usage.¹⁰⁶ Such models allow us to separate

¹⁰⁵ See ANGRIST & PISCHKE, *supra* note 54, at 15.

¹⁰⁶ See *id.* at 22–24 (explaining the use of OLS regression in experimental designs);

the effect attributable to AI-assistance from the effect of these other variables. The results are consistent with those reported in Table 2, and the average treatment effect does not meaningfully change across any of the four measures.¹⁰⁷

While these results demonstrate the substantial impact that AI has on the quality of the synthesis task, improvements in quality alone do not capture AI's full impact on performance. An equally important question is whether these improvements also enhance the participant's speed in completing the task. The *Synthesis Task Speed Hypothesis* predicts that participants in the AI-exposed group will complete the synthesis task in less time. To measure the speed at which participants completed the synthesis task, we asked participants to self-report the number of minutes it took to complete the task (Sample Mean = 64.69, Sample SD = 11.17).¹⁰⁸ Participants had a maximum of seventy-five minutes to complete the synthesis task.

TABLE 3: SYNTHESIS TASK: SPEED

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	Effect (% Change)	P-Value
Minutes	69.46 (7.41)	60.70 (12.25)	-8.76 (-12.62%)	<0.001***

Note: P-value estimated using a two-tailed t-test.

Table 3 reports the effects of AI assistance on the time it took, in total minutes, for participants to complete the synthesis task. The results support the *Synthesis Task Speed Hypothesis*. On average, participants in the control group completed the synthesis task in approximately seventy minutes, whereas those in the AI-exposed group completed it in sixty-one minutes. In other words, AI usage reduced the time spent on the synthesis task by nine minutes or 12.62%. Other studies have found similar effects, showing that AI

WOOLDRIDGE, *supra* note 99, at 78–81.

¹⁰⁷ See *infra* App. B.2.

¹⁰⁸ One participant did not complete this task and, therefore, is excluded from this analysis.

reduces the time required to complete tasks.¹⁰⁹

Our final measure of performance is an aggregate measure of productivity. Consistent with prior studies,¹¹⁰ we define productivity by dividing our measure of overall quality by the number of minutes it took the participant to complete the synthesis task (Sample Mean = 0.15, Sample SD = 0.07). Table 4 reports the effects of AI assistance on productivity. The results are consistent with our other hypotheses. On average, participants in the control group scored 0.10 points per minute, whereas those in the AI-exposed group scored 0.19 points per minute. In other words, AI usage improved overall productivity by 92.36%. We observe no meaningful difference when we estimate OLS models with controls for GPA, year in law school, and AI usage.¹¹¹

TABLE 4: SYNTHESIS TASK: PRODUCTIVITY

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	Effect (% Change)	P-Value
Productivity	0.10 (0.03)	0.19 (0.07)	+0.09 (+92.36%)	<0.001***

Note: P-value estimated using a two-tailed t-test.

One concern with the use of AI for legal tasks is that the AI may “hallucinate,” fabricating citations or mischaracterizing legal rules in ways that are difficult for users to detect.¹¹² We observed very few fabricated or misstated legal rules in either group. Participants in the control group and the AI-exposed group were equally likely to fabricate or misstate legal rules.¹¹³

¹⁰⁹ See Schwarcz et al., *supra* note 2, at 16 (observing a twenty to twenty-eight percent reduction in the time it takes to complete five legal tasks); Choi, Monahan, & Schwarcz, *supra* note 2, at 174 (observing a twelve to thirty-two percent reduction in the time it takes to complete four legal tasks).

¹¹⁰ See Schwarcz et al., *supra* note 2, at 17.

¹¹¹ See *infra* App. B.2.

¹¹² See *infra* Part I.A.

¹¹³ We observed only one notable mischaracterization across the synthesis tasks. A small number of participants in both the control group (N = 3; 7.3%) and the AI-assisted group (N = 2; 6.0%) invented a “sufficiency” standard for the notice requirement, even though the materials contained no such rule. We find no evidence that participants in the AI-assisted condition were more prone to this error. One plausible explanation is that many doctrines in property and contract law evaluate the sufficiency of notice, leading students to import

The absence of hallucinations in our study should not be interpreted as evidence that AI models no longer hallucinate. Rather, our results show that AI outputs can be reliable for summarizing a closed set of documents when properly supervised.

B. Results for the Comprehension Task

The primary goal of our study is to assess whether the use of AI for initial lawyering tasks affects participants' subsequent ability to engage in independent legal reasoning when they no longer have access to AI. Consistent with the possibility that AI use may impair human comprehension of legal source material, the *Comprehension Hypothesis* predicts that participants in the AI-exposed group would receive lower scores on a six question, multiple-choice quiz designed to evaluate their comprehension of the material without the aid of AI (Sample Mean = 3.88, Sample SD = 1.28).

TABLE 5: COMPREHENSION TASK: NUMBER OF CORRECT QUESTIONS

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	Effect (% Change)	P-Value
Comprehension	3.86 (1.37)	3.88 (1.21)	+0.02 (+0.57%)	0.935

Note: P-value estimated using a two-tailed t-test.

The data in Table 5 are not consistent with this hypothesis, showing no significant difference in the ability of participants to comprehend information from the source packet. On average, participants in the control group scored 3.86 points, whereas those in the AI-exposed group scored 3.88 points. At least fifty percent of the participants answered each question correctly, and no question was answered correctly by more than eighty percent of participants, supporting the validity of the comprehension test.¹¹⁴ Breaking

familiar concepts into their memoranda.

¹¹⁴ If the questions were too easy, nearly all participants in both groups would answer correctly, creating a ceiling effect that would mask any true difference in comprehension. Conversely, if the questions were too difficult, floor effects would similarly prevent detection of meaningful differences. The observed range of 50% to 80% correct responses indicates that the questions possessed sufficient discriminative power to reveal differences in

down the results by question does not meaningfully change the result.¹¹⁵ These findings therefore provide no evidence that AI assistance undermines participants' comprehension of the underlying legal materials when comprehension is assessed soon after exposure to the materials.

C. Results for the Application Task

The second skill necessary for legal reasoning is the ability to apply legal principles to factual situations.¹¹⁶ The *Application Task Quilty Hypothesis* predicts that participants who use AI to distill the key legal principles from the relevant source materials will perform worse than the control group on the subsequent application task that requires them to apply those principles without AI assistance. Again, we evaluate performance on the application task across three dimensions: quality, speed, and productivity. To measure quality, each memo was graded along three dimensions:

- *Focus and Emphasis* (Scored: 1-5): Overall impression of whether the memo relies on the strongest arguments regarding the most important issues, as opposed to addressing tangents or weak points. (Sample Mean = 1.96, Sample SD = 1.06).¹¹⁷

comprehension between groups, had such differences existed. See Frederic M. Lord, *The Relation of the Reliability of Multiple-Choice Tests to the Distribution of Item Difficulties*, 17 PSYCHOMETRIKA 181, 190 (1952) (mathematically proving that test reliability is maximized “when the item difficulty is somewhat easier than halfway between the change level and 1.00.”). In our case, the ideal item difficulty would be approximately 62.5%.

¹¹⁵ See *infra* App. D. The AI-assisted group outperformed the control group on one question, while the control group outperformed the AI-assisted group on another. We have no hypothesis to explain why the differences arose on these particular questions and, therefore, the best explanation is that the statistical significance is due to random chance rather than meaningful differentiation.

¹¹⁶ See *supra* part II.A.1.

¹¹⁷ Scores on the application task were generally low, with a sample mean of 5.99 out of a possible 15 points (SD = 2.61). On two subdimensions, the modal score was 1, and a substantial share of observations were clustered near the lower bound. This pattern is consistent with, though not definitive evidence of, a floor effect, in which the scale's lower limit compresses variation among lower-performing participants. Floor effects can bias estimated treatment effects by reducing the observable distance between individuals whose latent scores differ below the minimum of the scale.

To formalize the issue, let Y^* denote a participant's latent ability on the application task, and let the observed score be $Y = \max(Y^*, 1)$. For participants whose latent scores fall below 1, the observed score is censored at $Y = 1$ even though $Y^* < 1$. If censoring differs across experimental conditions, comparisons of observed means may not accurately reflect differences in latent ability, and the direction of any resulting bias depends on how censoring

- *Depth of Analysis and Understanding* (Scored: 1-5): Does the memo merely assert arguments favoring a position, or does it develop arguments, including by integrating the record, precedent, and creative reasoning in a manner that suggests understanding of the issue? (Sample Mean = 1.73, Sample SD = 0.98).
- *Organization* (Scored: 1-5): Overall impression of whether the memo organizes the material in an effective and reasonable matter. (Sample Mean = 2.30, Sample SD = 0.94).¹¹⁸

We added these three components to produce a composite measure of overall quality (Sample Mean = 5.99, Sample SD = 2.61). To estimate the average treatment effect, we use a difference-in-means to assess whether the AI-exposed group and the control group differed in their performance on the application task.

is distributed across groups. If control participants are more likely to be censored at the floor, this would bias the treatment effect downward. If the AI-exposed participants are more likely to be censored at the floor, this would bias the treatment effect upward. Given our finding that the AI-exposed group outperformed the control group, the latter scenario would change our interpretation of the results by making it more likely that there is no difference between the two groups.

The available summary statistics tend to suggest that the lowest scores were more common in the control group. On *Overall Quality*, ten participants (24.4%) in the control group received the lowest possible score, compared to four (8.0%) in the AI-exposed group. On *Focus and Emphasis*, twenty-one participants (51.2%) in the control group received the lowest possible score, compared to nineteen (38.0%) in the AI-exposed group. On *Depth of Analysis and Understanding*, twenty-seven participants (65.9%) in the control group received the lowest possible score, compared to twenty-one (42.0%) in the AI-exposed group. Taken together, these patterns provide some evidence that the control group is more likely to experience the sort of censoring associated with floor effects.

As a robustness check, we estimate a Tobit model, which treats the outcome as a censored realization of an underlying continuous latent variable. The estimated effect of the AI-exposed treatment is significant and positive ($\beta = 1.27, p = 0.017$) on the latent scale, indicating that the treatment is associated with higher underlying application performance under the model's assumptions. Accordingly, although the score distribution is consistent with some compression near the floor, the principal results do not appear to be driven by lower-bound censoring.

¹¹⁸ We did code for “polish” but did not include it in our overall score for this task because we explicitly instructed students to focus on the substance of their memos rather than writing conventions.

TABLE 6: APPLICATION TASK: QUALITY

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	Effect (% Change)	P-Value
Overall Quality	5.29 (2.50)	6.56 (2.58)	+1.27 (+24.01%)	0.02*
Focus and Emphasis	1.76 (1.02)	2.12 (1.08)	+0.36 (+20.45%)	0.10
Depth of Analysis	1.51 (0.87)	1.92 (1.03)	+0.41 (+27.15%)	0.04*
Organization	2.02 (0.99)	2.52 (0.84)	+0.50 (+24.75%)	0.01*

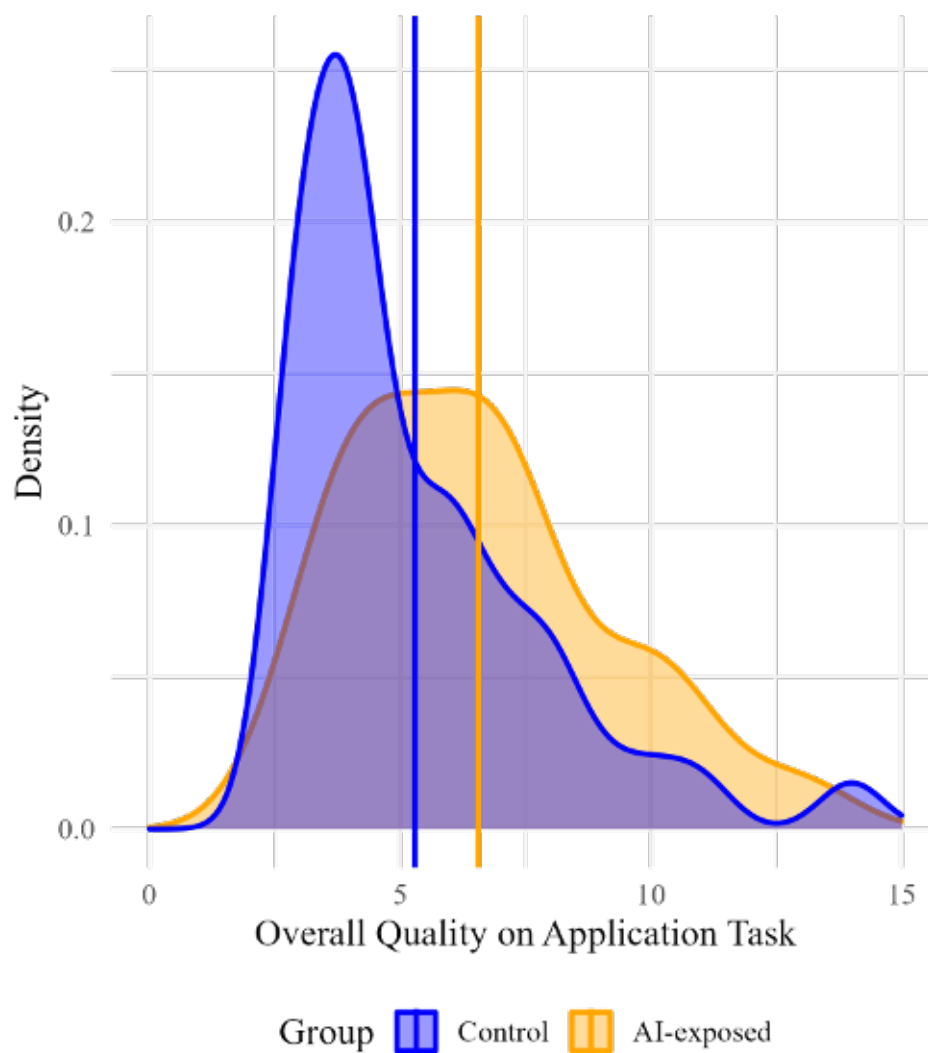
Note: P-values estimated using a two-tailed t-test.

Table 6 presents the quality of the reasoning memos for each group. Figure 3 depicts the simple distribution of the overall quality on the synthesis task. Contrary to the *Application Task Quality Hypothesis*, participants in the AI-exposed group scored significantly *higher* than participants in the control group even after they no longer had access to AI. On average, participants in the control group scored 5.29 points, whereas those in the AI-exposed group scored 6.56 points. In other words, AI use on the initial synthesis task improved overall performance on the later application task by 24.01%. Participants in the AI-exposed group outperformed the control group across all three measures incorporated into the measure of overall performance. Estimates attained from OLS regression models with controls for the student's GPA, year, in law school, and prior AI usage do not meaningfully change the results.¹¹⁹

The effect sizes illustrate the significance in the difference between participants in the AI-exposed group and those in the control group. We observe an increase of nearly half a standard deviation (0.49 standard deviations) in overall performance on the application task. Framed differently, the average participant in the control group scored in the 39th percentile. By contrast, the average participant in the AI-exposed group scored in the 59th percentile.

¹¹⁹ See *infra* App. E.2.

FIGURE 2: OVERALL SCORE DISTRIBUTION ON APPLICATION TASK



These findings raise significant questions about the mechanism that may have enabled participants in the AI-exposed group to outperform participants in the control group. Because no participant had access to AI for the application task, performance differences cannot be attributed to real-time AI assistance. One possible explanation is that participants in the AI-exposed group produced stronger synthesis memos,¹²⁰ which in turn improved their

¹²⁰ See *supra* Part III.A.

ability to complete the application task. We test this possibility by estimating OLS models controlling for each participant's performance on the synthesis task.

TABLE 7: PERFORMANCE ON APPLICATION TASK WITH CONTROL FOR PERFORMANCE ON SYNTHESIS TASK

	Overall Quality
AI-exposed	0.56 (0.63)
Synthesis Task Score	0.23* (0.09)
GPA	1.10 (0.72)
3L	-0.01 (0.50)
Constant	-0.26 (2.58)
AI Usage Control	Yes
N	91
Estimator	OLS

Note: * $p < .05$; ** $p < .01$; *** $p < .001$

Table 7 presents the estimates from this model. Once we control for participants' performance on the synthesis task, the AI-exposed treatment is no longer statistically significant. The only significant predictor of performance on the application task is performance on the synthesis task. A one standard deviation increase (3.30) in synthesis-task performance is associated with a 0.76 point increase on the application task. In other words, participants in the AI-exposed group produced higher quality reasoning memos only to the extent that AI helped them produce stronger synthesis memos in the first instance. These results suggest that AI's ability to improve subsequent legal reasoning operates indirectly, by improving the quality of initial work on which subsequent work builds.

One possible consequence of AI use on synthesis tasks, however, is that it increases the time that attorneys spend completing later application tasks when they no longer have access to AI. Accordingly, the *Application Task*

Speed Hypothesis predicts that the AI-exposed group will take longer to complete the application task compared to the control group.

TABLE 8: APPLICATION TASK: SPEED

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	Effect (% Change)	P-Value
Minutes	49.63 (10.32)	53.13 (8.50)	+3.50 (+7.10%)	0.09

Note: P-value estimated using a two-tailed t-test.

Table 8 reports the effects of AI assistance on the time participants devoted to the application task. Although the results are directionally consistent with our hypothesis,¹²¹ they do not reach the threshold of statistical significance. On average, participants in the AI-exposed group took longer to draft their application memos than participants in the control group. One possible explanation is that AI-exposed participants needed to consult the source packet more frequently while composing the reasoning memo. Yet the comparable performance of the two groups on the comprehension task suggests that AI-exposed participants did not differ meaningfully from control participants in their ability to recall the relevant material.

Unlike the synthesis task, we observe at least some trade-offs between quality and speed for purposes of legal reasoning. Although participants in the AI-exposed group produced higher quality memos, they took slightly longer to complete the task. To evaluate these trade-offs, we compare the productivity of the two groups. Again, we measure productivity on the application task by dividing overall quality by the time it took the participant to complete the task. Table 9 reports the effects of AI assistance on the points scored per minute. We find no statistically significant difference in productivity between participants in the AI-exposed group and those in the

¹²¹ See *supra* Part I.C.

control group.

TABLE 9: APPLICATION TASK: PRODUCTIVITY

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	Effect (% Change)	P-Value
Productivity	0.11 (0.05)	0.12 (0.05)	+0.01 (+13.20%)	0.202

Note: P-value estimated using a two-tailed t-test.

One should not read these results, however, to suggest that AI offers no improvements in performance. Between the synthesis and application tasks, participants in the AI-exposed group completed the whole study an average of 5.26 minutes faster than participants in the control group.¹²² Although both groups performed similarly on the application task as measured by productivity, participants in the AI-exposed group still produced higher quality memos overall and completed the synthesis task in significantly less time.

D. Results for the Revision Task

Although the study primarily examined the downstream effects of using AI to synthesize legal source material, it also tested the impact of using AI to revise work originally produced without AI assistance.¹²³ To do so, the final task instructed all participants to use AI to revise their application memos, which had been written without the assistance of AI. The *Revision Hypothesis* predicts that access to AI will help all participants improve the quality of their work, but that participants who started with weaker memos will see a larger improvement than participants who started with stronger memos. We measured the quality of each revised memo using the same criteria that we used in the application task.

As Figure 3 illustrates, we find partial support for the *Revision Hypothesis*. Participants whose initial memos were relatively weak produced stronger memos after revising with AI. Surprisingly, however, participants

¹²² Compare Table 3, *supra*, with Table 8, *supra*.

¹²³ See *supra* Part II.A.2.d.

whose initial memos were relatively strong produced weaker memos after revising with AI. Because the slope appears to change around the mean, we estimate a piecewise linear regression model to determine how the magnitude of the decline varies across participants. A piecewise model allows the relationship between initial performance and the change in score to differ above and below the mean while still treating the original score as a continuous variable.

FIGURE 3: CHANGE IN OVERALL QUALITY

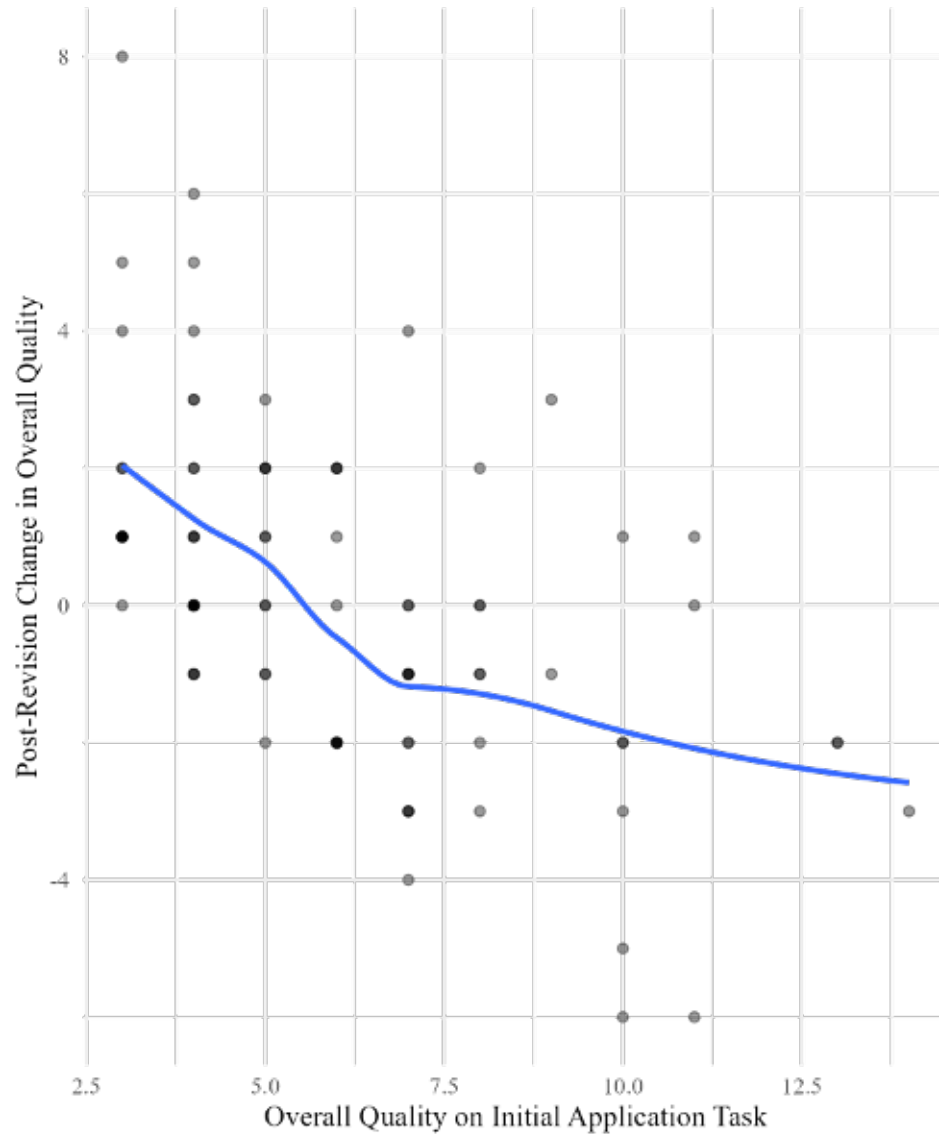


Table 10 reports the results of the regression, confirming the pattern illustrated by Figure 3. Participants who scored between three and five points on the application task generally improved during the revision task. For example, the lowest-scoring participants (i.e., those who scored 3 points) experienced an estimated 1.91-point increase on the revision task. By contrast, participants who scored five points or higher on the application task

experienced a decline during the revision task. The highest-scoring participants (those who scored fourteen points) experienced an estimated 8.08-point decrease on the revision task. Notably, there is no statistically distinguishable difference on this task between participants in the AI-exposed group and the control group. The results hold if we examine the change in depth, focus, or organization instead. The results do not meaningfully change if we control for GPA, year in law school, or AI usage.

TABLE 10. REVISION TASK: CHANGE IN OVERALL QUALITY

	Change in Overall Score
Overall Score on Application task	-0.97*** (0.22)
Above the Mean on Application task	0.68* (0.31)
AI Exposure	0.70 (0.44)
Constant	4.82*** (1.01)
AI Usage Control	Yes
N	91
Estimator	OLS

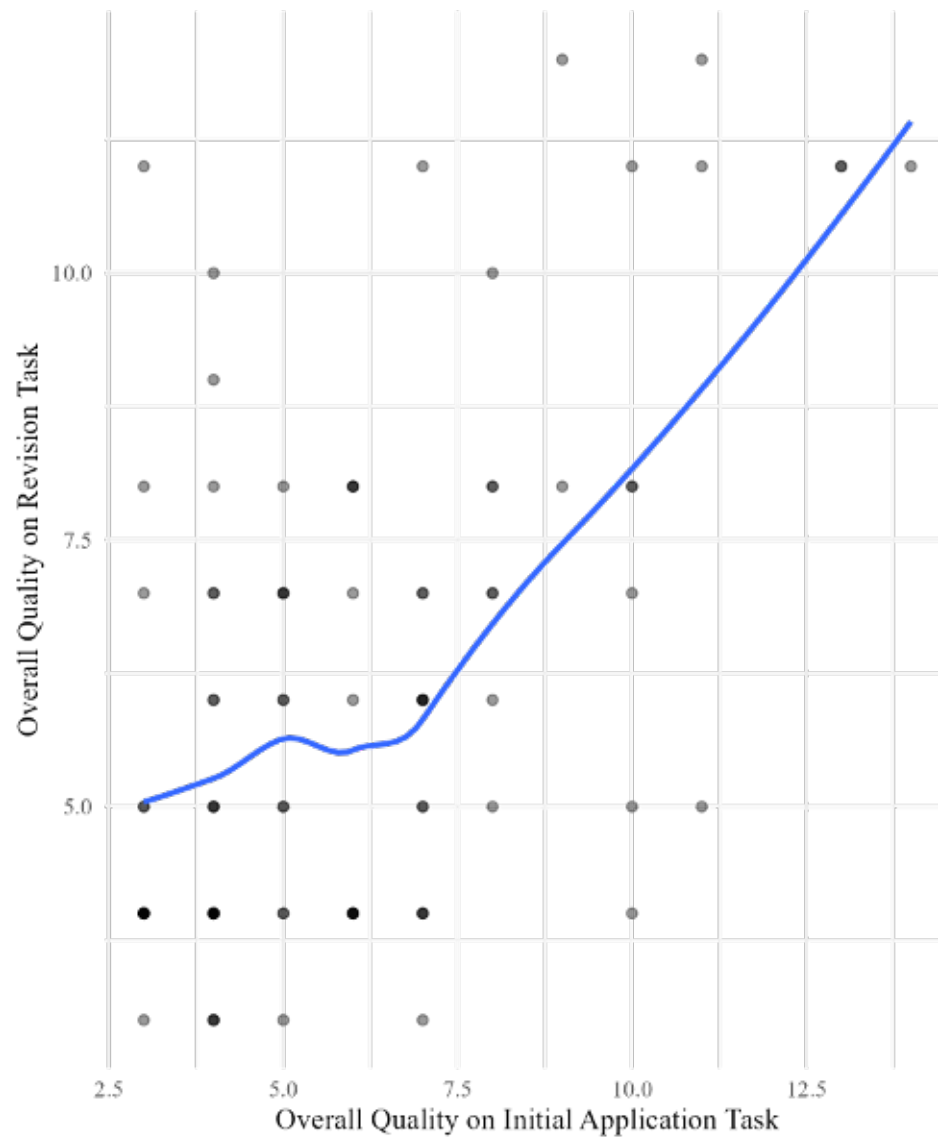
Note: * $p < .05$; ** $p < .01$; *** $p < .001$

It is important to emphasize that the *Revision Hypothesis* relates to the *change* to an individual participant's performance—not their raw performance on the revision task.¹²⁴ As Figure 4 demonstrates, participants who wrote a strong initial memo produced revised memos that were stronger than the revised memos of participants whose initial memos were weak. In other words, although AI-exposed revision tended to reduce the scores of the highest-performing participants, it did not reverse the overall ranking of participants. High-performing participants continued to produce stronger work than lower-performing participants even after revision. The results therefore suggest that AI-exposed revision may compress differences in performance—improving weaker work while slightly degrading stronger

¹²⁴ See *supra* Table 1.

work—without eliminating significant gaps in the quality of the underlying legal reasoning.

FIGURE 4. RELATIONSHIP BETWEEN OVERALL QUALITY ON APPLICATION TASK AND REVISION TASK



Our study does not permit a thorough investigation into why high-performing participants experienced a decline in the quality of their

memos.¹²⁵ One possible explanation is that AI produces a leveling effect by improving weaker drafts while standardizing or flattening stronger ones. Participants who produced sophisticated, well-structured analyses during the application task may have incorporated AI-generated edits that simplified, reformulated, or reorganized their arguments in ways that reduced nuance, precision, or doctrinal depth.¹²⁶ Participants who initially produced strong memos may have failed to recognize these problematic edits given the time constraints associated with this task and the cognitive fatigue they experienced towards the end of the three-hour experiment, which may have increased their willingness to defer to the AI-generated output.¹²⁷ Alternatively, participants may have deferred to the AI's suggestions, failing to critically consider whether they meaningfully improved the overall memo.

Existing research is consistent with this explanation. At least one study has found that the highest-performing participants on a law-school essay experienced “moderate and statistically significant declines when given access to AI.”¹²⁸ Other studies likewise report that AI usage tends to improve performance on lawyering tasks for the lowest performers while providing no benefit to the highest performers.¹²⁹ Our study contributes to this growing body of evidence suggesting that AI usage by high-performing individuals may degrade work product. At the same time, we cannot determine with confidence whether this effect reflects AI leveling work product, cognitive fatigue, unwarranted trust in AI models, or a mixture of these possible

¹²⁵ One possible explanation for this pattern is that higher-performing participants may have spent less time carefully reviewing and editing the AI-generated revisions. Our data, however, do not support this possibility. Participants who produced higher-quality memos on the application task spent more time on the revision task. We have no theoretical reason to believe that spending more time on the revision task would lead participants to degrade the quality of their memos.

¹²⁶ Consistent with this explanation, empirical research suggests that AI revisions often alter the meaning of text, notwithstanding user prompts instructing the AI not to do so. *See* Marwa Abdulhai et al., *How LLMs Distort Our Written Language*, arXiv:2603.18161 (2026).

¹²⁷ *See* VIRGINIA EUBANKS, *AUTOMATING INEQUALITY: HOW HIGH-TECH TOOLS PROFILE, POLICE, AND PUNISH THE POOR* (2018) (discussing the reluctance of many humans to override automated decision-making).

¹²⁸ *See* Jonathan H. Choi & Daniel Schwarcz, *AI Assistance in Legal Analysis: An Empirical Study*, 73 J. LEGAL EDUC. 384, 401 (2025).

¹²⁹ *See* Choi, Monahan, & Schwarcz, *supra* note 2, at 181 (examining performance on four tasks: complaint drafting, contract drafting, drafting an employee handbook, and a client memo).

explanations.

E. External Validity and Other Empirical Limitations

Although our findings provide important initial evidence regarding the short-term effects of AI use on legal reasoning, several limitations qualify the strength and generalizability of our conclusions. As described above, the study was conducted with upper-level law students at a single institution and focused on a specific doctrinal area—servitudes on personal property—within a tightly controlled setting. While this design enhanced internal validity and allowed us to isolate the causal effect of AI on legal reasoning, it necessarily constrains external validity.

First, law students differ from practicing lawyers in experience, legal knowledge, professional incentives, and exposure to real-world consequences. Newly licensed lawyers may resemble our participants in some respects, but seasoned practitioners may interact with AI differently—either relying on it more strategically or resisting it based on established habits and expertise. In addition, practitioners may use AI in response to meaningful time pressure, whereas law students responding to a study by faculty members may be inclined to be more diligent because their time commitment is fixed, capped, and limited to the assigned tasks. Consequently, the cognitive behavior of these two groups may differ in ways that ultimately affect performance.

Second, selection effects in the real world may introduce complications that our controlled study excludes. In the real world, lawyers use AI only when they want to. In our study, we randomly required certain participants to use AI. It is possible that lawyers who choose to use AI may not share the same characteristics as the participants who were forced to use AI. For example, judicial opinions sanctioning lawyers who filed briefs with hallucinated citations often document a disturbing lack of diligence and professional responsibility.¹³⁰ If one surmises from these examples that AI

¹³⁰ See Pamela Langham, *U.S. District Court of Maryland Addresses Hallucinated Cases*, MD. ST. BAR ASS'N (Jan. 7, 2026), <https://www.msba.org/site/site/content/News-and-Publications/News/General-News/U.S.%20District%20Court%20of%20Maryland%20Addresses%20Hallucinated%20Cases.aspx> (“In two separate matters, although involving the same lawyer, the court confronted legal pleadings in which counsel relied on non-existent authorities cited in the court filings, raising serious questions about diligence, ethics, and the integration of AI

may be especially attractive to lawyers who are pressed for time and willing to skimp on diligence, then the negative consequences of AI use might be more significant—at least for that population—than it was for the participants in our study who did not use AI by choice.¹³¹

Third, the study examines only short-term effects within a single three-hour session. This design cannot capture the long-term cognitive consequences of repeated AI reliance over months or years. Skill erosion, if it occurs, is likely to be cumulative and gradual rather than immediate. Conversely, repeated exposure to AI may allow individuals to use it more skillfully, improving their overall ability to synthesize legal materials and apply them to specific fact patterns.

Fourth, the experiment involved a closed universe of limited legal materials in a hypothetical jurisdiction. Real-world legal practice often involves open-ended research with numerous ambiguous authorities. AI may be good at synthesizing a limited set of curated materials, but may overlook key information or hallucinate when asked to identify relevant materials from a massive non-curated database, such as those available through Westlaw or LexisNexis. Accordingly, our results only speak to the efficacy of a human-in-the-loop model of AI use involving a limited set of relevant materials.¹³²

Fifth, we required participants in the AI-exposed group to use a specific AI model (Gemini 2.5 Pro) under structured prompting instructions. Results may differ with different foundation models, models developed specifically for legal use, different prompting strategies, or unstructured AI use.

Finally, our experimental design required us to fragment a standard legal research assignment into an unrealistically narrow set of component tasks, consisting of reading and synthesizing specific legal materials, applying them to a specific hypothetical, and revising an earlier draft. This is not typical of how most lawyers structure assignments. In practice, partners and senior attorneys usually assign junior lawyers tasks that require consideration of

technologies into legal research and drafting.”).

¹³¹ Of course, AI might also be attractive even to diligent lawyers who view it as a valuable addition to their professional toolkit. The point is simply that our results show how a random sample of participants use AI, not how a self-selected group of lawyers would use it.

¹³² Practicing attorneys may face similar constraints when required to absorb large quantities of information quickly. We cannot determine whether there is a threshold at which marginal gains in efficiency begin to produce diminishing returns in the accuracy of synthesis tasks.

more expansive and less concrete questions, and they rarely supply attorneys with a narrow and closed universe of potentially relevant source material. We structured the experiment in this way to facilitate testing how AI use on an initial task would affect performance on a later, related task. But this structure inevitably limits the realism of the discrete lawyering tasks that our participants completed.

For these reasons, our results should be understood as evidence about a specific population, task type, time horizon, and technological configuration. The results meaningfully inform the debate over whether AI use inevitably undermines legal reasoning in the short term. But they do not resolve broader questions about whether sustained AI use erodes legal-reasoning skills or when short-term use of AI risks undermining individual legal reasoning across different tasks or practice settings.

IV. Implications

A. *Paths for Future Research*

As Part I explains, a growing body of evidence indicates that young lawyers and law students who use AI effectively produce higher-quality work in less time than those who do not.¹³³ Our results not only reinforce this conclusion, but suggest that these benefits of AI usage can persist after AI is no longer available, enhancing participants' application of governing legal rules, precedents, and doctrinal nuance. At the same time, our findings underscore meaningful risks of AI usage for lawyering tasks, as performance deteriorated for participants with strong initial memos once they were permitted to use AI for revision.¹³⁴

This constellation of findings suggests the need for additional research on how and when AI usage impacts human legal reasoning. Future studies can address some of the limitations we identified in Part III.E. A central priority for future research should involve additional randomized controlled trials, given their unmatched ability to identify causal mechanisms.¹³⁵ One

¹³³ See *supra* notes 25–32 and accompanying text.

¹³⁴ See *supra* Part III.D.

¹³⁵ See Rebecca C. Richmond, *Approaches for Drawing Causal Inferences from Epidemiological Birth Cohorts: A Review*, 90 EARLY HUM. DEV. 769, 770–71 (2014) (referring to large randomized controlled trials as the “gold standard” for estimating causal effects in the context of population health).

straightforward extension would replicate our core experimental structure while varying the nature of the underlying legal tasks, the time constraints under which participants operate, and the incentive structures that shape their effort and engagement. Such studies could help determine whether the effects we observe are robust across doctrinal domains, task complexity, and performance pressures. These studies could also provide evidence about why our findings departed from our predicted hypothesis for both the application and the revision hypotheses.¹³⁶ Future randomized controlled trials might also focus on practicing attorneys rather than upper-level law students.¹³⁷ Yet another approach might be to use participants without any legal training to better understand the relationship between AI tools and increasing rates of pro se litigation.¹³⁸

Although randomized controlled trials promote internal validity and

¹³⁶ Because our findings departed from our predicted hypothesis, additional studies are necessary to explain the precise mechanism for *why* participants in the AI-exposed group outperformed those in the control. Other randomized controlled studies might modify key elements of our design to better understand the mechanism. For example, one could attempt to neutralize the instrumental advantages of early-stage AI use by providing all participants, at a later stage, with a standardized synthesis of the governing law. Doing so would help isolate the cognitive effects of AI assistance in the initial phase from the downstream benefits that stem simply from having produced a stronger foundational work product. More ambitiously, future randomized controlled trials could seek to evaluate the longer-term consequences of AI-assisted legal reasoning by requiring participants to complete tasks with week-long or month-long gaps between each task. Another approach could isolate concerns that AI is attractive in part for its speed, which is a double-edged sword in a time-pressured practice environment. Cf. Liz Mineo & Dan Levy, *Is AI dulling our minds?*, HARV. GAZETTE (Nov. 13, 2025), <https://news.harvard.edu/gazette/story/2025/11/is-ai-dulling-our-minds> (“AI can also hinder learning when students are overcommitted, overworked, and see AI exclusively as a time-saving device.”). A study might provide three groups with a combined synthesis and application task. One group would have a short time limit and could use AI; the second group would have a longer time limit and could not use AI; and third group would have both the longer time limit and access to AI. Additional research is also necessary to understand why high-performing participants saw a decrease in performance during the revision task. Subsequent research might focus on this question by instructing participants to use AI in an application task, and comparing the quality of AI-supported output between the initial application task and the revision task.

¹³⁷ Although such studies present meaningful financial challenges, they may be feasible at a manageable cost by relying on contract attorneys. The substantial investments currently being made to understand how AI is reshaping legal practice suggest that external funding for this type of experiment may be attainable. Alternatively, researchers may be able to recruit a sufficiently large pool of practicing attorneys with only modest compensation. Either way, questions about representativeness would remain, particularly regarding differences between contract lawyers and attorneys at other levels of seniority.

¹³⁸ See, e.g., Mostafa Soliman, *Pro Se Advocacy in the AI Era: Benefits, Challenges, and*

enable testing of causal mechanisms, we are cognizant of the limitations of such studies.¹³⁹ A comprehensive research agenda should therefore incorporate alternative research design, such as field experiments and longitudinal observational studies in professional legal settings. Unfortunately, such research faces substantial practical and ethical challenges given the sensitivity of the data and the risk that such work might reveal risks to clients that practitioners themselves do not fully perceive or disclose.¹⁴⁰ Observational research is also crucial for evaluating the long-term effects of AI use on the development of lawyering skills, although isolating selection effects and accurately measuring distinct patterns of AI use will be challenging.

B. Suggestions for Best Practices

The risk that AI will undermine human users' independent legal thinking is real, but it is neither inevitable nor unmanageable. The key to managing the threat to independent legal thinking lies in confining AI use to settings where displacement risk is minimal. Perhaps the most significant such domain involves tasks where the attorney can reliably assess, adapt, explain, and build upon the AI's output from her own experience and knowledge. Many routine legal tasks satisfy this standard. Skilled writers can use AI to improve the clarity and flow of a paragraph they have drafted, precisely because their own editorial skills allow them to evaluate and further refine what the tool produces.¹⁴¹ An experienced transactional lawyer can use AI to

Ethical Implications, N.Y. St. Bar Ass'n (Feb. 10, 2026), <https://nysba.org/pro-se-advocacy-in-the-ai-era-benefits-challenges-and-ethical-implications/>

¹³⁹ See *supra* Part III.E.

¹⁴⁰ One pathway for overcoming these challenges might be to study the effects of AI usage in law school clinics or nonprofit legal service providers. However, such studies might raise ethical problems, particularly if they involved a randomization of which clients received AI-facilitated legal services and others did not. Another possibility is collaboration with non-legal companies that are integrating AI into their in-house legal departments. Because these entities are not selling legal services in a competitive professional marketplace, they may face fewer external market penalties if research identifies limitations or risks in current AI practices.

¹⁴¹ See *The Role of AI in Academic Editing: What Editors Need to Know*, WILEY (Sept. 2025), <https://editors.wiley.com/page/the-role-of-ai-in-academic-editing> ("Using AI effectively requires the ability to both guide its tasks and critically assess its output. Your expertise remains essential in determining what is relevant, credible, and valuable to the scholarly conversation.").

suggest supplemental contractual language or flag potential ambiguities in a familiar deal type, because she is well-positioned to evaluate those suggestions and iterate on them to maximize their usefulness.¹⁴² Lawyers can also use AI to simplify documents they already understand for the purpose of communicating with clients or junior colleagues, or to adapt a standard template to a modestly different factual context.¹⁴³

The foregoing examples resonate with how law firms have historically operated. Senior lawyers drafting documents frequently receive suggestions from junior colleagues. These senior lawyers exercise judgment when deciding which suggestions to accept and which to reject because senior lawyers know that junior lawyers sometimes make mistakes. A similar approach is warranted for AI. Senior lawyers must learn to exercise judgment regarding when and how to integrate suggestions from AI, rather than treating AI as infallible. Junior lawyers must develop the same critical approach. Law firms that for decades have trained their lawyers to assess each others' advice should be able to develop similar training for assessing advice from AI.

By contrast, lawyers who deploy AI to generate work product they cannot fully understand, question, explain, or modify risk displacing the hard cognitive labor that professional competence requires.¹⁴⁴ Understanding new and complex legal material is not simply a matter of receiving or transmitting correct information; it is the product of a demanding mental process that cannot be outsourced without compromising legal judgment. A lawyer who relies on AI to construct an unfamiliar legal argument, resolve a complex research question, or explain the significance of an unread case or contract clause may obtain output that appears professionally adequate while remaining personally unable to defend, refine, or accurately represent it.¹⁴⁵ But over time such lawyers will fail to make connections across discrete tasks and related cases. They will stumble to explain their work to partners, clients,

¹⁴² See Zack Shapiro, *The Claude-Native Law Firm*, X (Feb. 27, 2026), <https://x.com/zackbshapiro/status/2027389987444957625>.

¹⁴³ See *id.*

¹⁴⁴ Cf. Angxuan Chen et al., *Learning by Teaching with ChatGPT: The Effect of Teachable ChatGPT Agent on Programming Education*, 57 BRIT. J. EDUC. TECH. 163 (2026) ("Findings indicate that teaching ChatGPT improved students' knowledge gains and programming abilities However, its impact on error-correction skills was limited, likely due to ChatGPT's tendency to generate correct code, thus reducing debugging opportunities.").

¹⁴⁵ See *supra* Part I.B.

or judges. They will be ill equipped to adapt their work to new settings. And, in the long-term, they will never develop the critical legal thinking skills and professional judgement that expert lawyering requires.

Newly licensed lawyers are particularly susceptible to this failure mode, and the consequences for their professional development are especially severe.¹⁴⁶ The analytical habits that define legal competence—learning to sit with a hard question, to trace an argument through its premises, to recognize when doctrine is uncertain and when it is settled—are built through repeated engagement with difficult material, not through efficient delegation of that material to a machine.¹⁴⁷ However, senior lawyers are not immune to this dynamic, even if they are less acutely vulnerable to it. An experienced partner who uses AI to reason through an unfamiliar area of law may face the same structural problem as the newly licensed lawyer, albeit perhaps with greater awareness of her own limitations.

A second essential principle for using AI without undermining the human user's independent legal reasoning is to employ it for narrow, well-defined tasks rather than for broad assignments that bundle together multiple analytical steps.¹⁴⁸ When AI is asked to complete an entire complex project, such as drafting a brief or revising a contract, it risks substituting for the lawyer's own reasoning across the full scope of the work. Less risky uses of AI involve clearly bounded components that allow the lawyer to retain responsibility for the structure and substance of the analysis.¹⁴⁹ For example, lawyers should not ask AI systems to draft a full brief or even an entire section. They can, however, use AI to help refine or generate individual paragraphs once the argument and legal theory are already established.¹⁵⁰ Likewise, instead of directing AI to produce or adapt an entire contract, lawyers can use it to examine, revise, or extend particular clauses, or to help

¹⁴⁶ See Joanne Harris, *AI Presents both Opportunity and Threat to Young Lawyers*, INTERNATIONAL BAR ASS'N (Jan. 7, 2026), <https://www.ibanet.org/AI-presents-both-opportunity-and-threat-to-young-lawyers>.

¹⁴⁷ See Schwarcz et al., *supra* note 7 (manuscript at 26–27); Choi, Monahan & Schwarcz, *supra* note 2, at 205–06 (noting the risks posed by providing access to full AI tools to students prior to them developing a foundation of legal and analytical skills).

¹⁴⁸ See Joe Regalia, *From Briefs to Bytes: How Generative AI is Transforming Legal Writing and Practice*, 59 TULSA L. REV. 193, 221 (2024).

¹⁴⁹ See *id.* at 214–16.

¹⁵⁰ See *id.*

locate specific provisions within a document.¹⁵¹

Like the principle that attorneys should rely on AI only when they can confidently assess, explain, and build upon its output, this second principle is designed to confine AI use to contexts where the risk of displacing independent legal reasoning is limited. Breaking broad legal problems into smaller tasks is itself a central part of legal analysis.¹⁵² Lawyers often discover new issues, complications, or counterarguments only after they begin to map the problem on their own terms and determine how the work should be structured. A lawyer may initially believe that they can fully understand and explain an AI-generated answer, consistent with the first principle, yet later recognize that the output overlooked an important issue, counterargument, or framing of the problem. By first using their own judgment to divide a legal assignment into discrete components, lawyers perform the cognitive work that reveals the relevant terrain of the problem. AI can then assist with particular steps within that framework, while the lawyer remains responsible for identifying the issues, structuring the analysis, and determining how the pieces fit together.

This principle may help explain the results of our experiment. The design of the study required participants to complete a relatively straightforward legal assignment, drafting a memorandum in response to a research question, but it did so by separating the work into several constituent steps.¹⁵³ Participants were asked to synthesize the relevant legal principles, apply those principles to a fact pattern, and then revise and refine the resulting analysis. Even within the synthesis task, we structured AI use carefully so that participants engaged the tool in a relatively specific and constrained way, broadly consistent with the way lawyers are often advised to incorporate AI into their workflows. The AI was also fed a relatively limited set of documents rather than being asked to conduct independent research. By organizing the assignment into distinct tasks and subtasks, the experiment

¹⁵¹ See *Generative AI for Legal Professionals: Top Use Cases*, THOMSON REUTERS (May 13, 2025), <https://legal.thomsonreuters.com/blog/generative-ai-for-legal-professionals-top-use-cases-tri>.

¹⁵² See Melanie L. Oxhorn, *A Heuristic Approach to Solving Complex Litigation Problems*, 92 U. CIN. L. REV. 668, 695 (2025) (“[T]he problem in question should be reformulated or restructured into an equivalent problem that arguably has a recognized solution . . . by breaking it down into parts that likewise are more readily capable of resolution.”).

¹⁵³ See *supra* Part II.A.2.

may have reduced the likelihood that participants would rely on AI to complete the assignment in an undifferentiated way that bypassed the process of forming their own mental map of the governing legal principles.

A final principle suggested by our experimental results is that lawyers should avoid using AI to complete legal tasks under artificially tight time constraints or when cognitively fatigued. The reason is straightforward. Careful legal reasoning requires time and sustained cognitive effort, and people are naturally inclined to conserve that effort when circumstances permit. When time pressure is severe or brain power exhausted, the temptation to rely on AI as a substitute for independent analysis becomes particularly strong. In those conditions, AI is more likely to short circuit the very cognitive processes through which lawyers test assumptions, identify counterarguments, and refine their understanding of the legal issues at stake. Similarly, verification of AI-produced material is a fundamental part of effective and ethical AI use for both avoiding hallucinations and ensuring accurate legal analysis, and that verification takes time.¹⁵⁴

This dynamic may help explain an unexpected pattern in our experiment. Participants who initially performed relatively well on the application task often saw their performance decline when they were later given the opportunity to use AI to improve their work.¹⁵⁵ By that point in the study, they had already completed nearly three hours of cognitively demanding work, and their earlier performance suggests they had exerted substantial mental effort. They were then asked to use AI to revise their answers within an artificially short window of only twenty minutes. Under those conditions, participants may have been especially prone to defer to AI output rather than carefully interrogating and integrating it with their own reasoning. The result was not an improvement in performance, but in many cases a modest deterioration.

Taken together, these considerations suggest that the responsible integration of AI into legal practice requires more than technical competence with the tools themselves.¹⁵⁶ It requires lawyers to exercise honest self-assessment about the boundaries of their own expertise and to resist the

¹⁵⁴ See Mark H. Francis & Ashwini Jarral, A Legal Practitioner's Guide to AI and Hallucinations, at 7–8 (Jan. 2026), <https://nationalcenterforstatecourts.app.box.com/v/Legal-practitioner-guide-AI>.

¹⁵⁵ See *supra* Part III.D.

¹⁵⁶ See Regalia, *supra* note 148, at 219, 221.

temptation, which AI makes unusually seductive, of substituting apparently fluent output that can be produced in a matter of minutes for genuine understanding. A useful heuristic is to ask, before relying on AI-generated work product, whether the lawyer could defend that work product in a demanding conversation with a skeptical colleague or judge without further preparation. If the answer is no, the task has likely been delegated rather than assisted, and the professional and developmental costs of that delegation may outweigh its efficiency gains.

Conclusion

This Article counsels against both complacency and panic. Our experiment provides some of the first causal evidence that AI use in legal work does not inevitably erode independent legal reasoning: when used to assist an initial synthesis task, AI improved participants' intermediate work product and, in turn, their later reasoning performance even after AI was removed. But our results also show that AI is not an unalloyed good. When introduced at the revision stage, it improved the work of some participants while degrading the work of others, suggesting that AI can sometimes displace rather than sharpen human judgment. The lesson, then, is not that the legal profession should embrace or reject AI wholesale. It is that lawyers, judges, educators, and clients must distinguish between forms of AI use that scaffold human reasoning and forms that supplant it. The future of legal practice will turn less on whether lawyers use AI than on whether they structure their use in ways that preserve the human capacities on which sound legal judgment ultimately depends.

Appendices

APPENDIX A. TREATMENT ASSIGNMENT AND BALANCE

1. *Identification Strategy*

Our identification strategy relies on a randomized controlled trial (RCT) designed to estimate the causal effect of AI assistance on participants' performance. Participants were randomly assigned to either a treatment group with access to Gemini 2.5 Pro during the synthesis task or a control group that completed the synthesis task without AI assistance. Random assignment ensures that, in expectation, both observable characteristics (such as prior academic performance or experience) and unobservable characteristics (such as motivation or baseline ability) are balanced across the two groups.¹⁵⁷ We assigned participants to the treatment conditions using an R script that randomly assigned each participant to a given condition with a probability of 0.5.

This identification strategy depends on the Stable Unit Treatment Value Assumption (SUTVA).¹⁵⁸ SUTVA requires, first, that the treatment assigned to one participant must not affect the outcomes of another participant. This assumption would be violated if one participant's exposure to the AI-exposed treatment "spilled over" to a participant in the control group.¹⁵⁹ Such spillover could occur if a participant in the AI-exposed group shared their output or strategies with someone in the control group.

To minimize the risk of spillover effects, we instructed the participants to complete the study in a single sitting and to refrain from discussing the study with other potential participants. We also administered the study through two separate Canvas courses—one for each group—to ensure that participants could not accidentally access materials intended for the other condition. Participants in the control group were told that they were "prohibited from using any AI tools to help" them complete the task. By contrast, participants in the AI-exposed group were instructed to begin by uploading the source packet to Gemini 2.5 Pro to ensure that they meaningfully engaged with the

¹⁵⁷ See ANGRIST & PISCHKE, *supra* note 54, at 15.

¹⁵⁸ SCOTT CUNNINGHAM, CAUSAL INFERENCE: THE MIXTAPE 141–42 (2021).

¹⁵⁹ *Id.* at 142.

AI output.

Second, SUTVA requires that the treatment be well defined and consistently administered across subjects.¹⁶⁰ Violations of this assumption occur most commonly when participants in the treatment group receive different “doses” of the treatment. For example, this assumption would be violated in a medical trial if some patients received higher or lower doses of an experimental drug. Researchers typically promote compliance with this requirement by defining the treatment in a way that minimizes variation within the treatment group.

In our study, all participants in the AI-exposed group were instructed to begin the synthesis task by uploading the source packet to Gemini 2.5 Pro and using a provided prompt to generate a short memo from the packet. We could not—and did not—control whether participants in the AI-exposed group continued to use AI after this initial intervention. Accordingly, the treatment in our study is defined as the initial exposure to the AI-generated synthesis rather than the amount of time participants spent using Gemini 2.5 Pro during the synthesis task.

Because treatment was randomly assigned, the average treatment effect of AI assistance can be expressed as a simple difference in mean outcomes between the treatment and control groups.¹⁶¹ Let Y_i denote the outcome for participant i , and let AI_i be an indicator variable equal to 1 if the participant was assigned to the AI-exposed group and 0 if assigned to the control group.¹⁶² The average treatment effect (β) can therefore be written as:

$$\beta = E[Y_i \mid AI_i = 1] - E[Y_i \mid AI_i = 0]$$

This expression represents the difference in the expected outcome for participants assigned to the AI-exposed group and those assigned to the control group. Because assignment is random, the distribution of both observable and unobservable characteristics should be balanced across groups.¹⁶³ As a result, the difference in mean outcomes provides an unbiased estimate of the causal effect of AI assistance.

¹⁶⁰ *See id.*

¹⁶¹ *See* ANGRIST & PISCHKE, *supra* note 54, at 15.

¹⁶² For an explanation of the mathematical notation here, see generally CUNNINGHAM, *supra* note 158, at 126–34.

¹⁶³ *See id.*

2. *Enrollment Survey*

Prior to the start of the experiment, we provided each participant with an enrollment survey. The survey served two purposes. First, it obtained the participant's informed consent and the information necessary to facilitate payment for their participation. Second, it collected several background characteristics about each participant. We gathered this information to conduct robustness checks and to examine whether the effects of AI assistance varied across participants within the treatment group. None of this information was used to assign participants to the treatment groups.

Year in Law School (Categorical Variable).

The participant's current student status. Because we did not permit first-year or LLM students to participate, this variable takes one of four possible values:

- I am a 2L who started my law school education at University of Minnesota Law School.
- I am a 3L who started my law school education at University of Minnesota Law School.
- I am a 2L who transferred to University of Minnesota Law School.
- I am a 3L who transferred to University of Minnesota Law School.

Approximately 46.1% of participants were 3L students. Only three participants transferred to University of Minnesota Law School. For purposes of our analyses, we represent this variable as a binary indicator of whether the participant was a 3L student or not.

1L GPA (Continuous Variable)

The participant's 1L GPA on a 4.00 scale. The highest GPA possible at University of Minnesota Law School is 4.333. The mean grade for first-year courses must fall between 3.200 and 3.333.¹⁶⁴ The mean GPA among our participants was 3.40, which suggests we attracted students who performed

¹⁶⁴ See UNIVERSITY OF MINNESOTA LAW SCHOOL, ACADEMIC RULES 22 (Mar. 4, 2025), <https://law.umn.edu/sites/law.umn.edu/files/2025-08/AcademicRules.3.4.25.pdf>.

slightly better than average during their first year of law school.

AI Usage (Categorical Variable)

The frequency with which the participant uses AI. We asked participants the following question: “Approximately how many times have you used generative AI tools for law school or other law-related task in the past three months?” Participants provided one of five possible responses:

- Never in the last three months.
- Once or twice in the last three months.
- Once a month.
- Once a week.
- Once a day.

The model response was “Once or twice in the last three months.” Accordingly, the average participant had *some* exposure to using AI for legal tasks, but that exposure was relatively limited.

3. *Balance within Groups*

By design, a randomized control trial eliminates selection bias among the treatment groups.¹⁶⁵ In fact, seeking to “balance” the treatment groups based on observable characteristics risks that researchers will—intentionally or unintentionally—distribute participants in ways that erroneously produce more statistically significant results.¹⁶⁶ We did not redistribute participants to obtain more balanced treatment groups. We provide the following analysis of balance for two reasons. First, many empiricists expect descriptive information on the sample. Second, we experienced a degree of differential attrition, which is one of the few instances in which balance testing may be appropriate.¹⁶⁷

We recruited a total of 106 participants for the experiment. A total of 91 participants completed the experiment.¹⁶⁸ Thirteen participants withdrew

¹⁶⁵ See ANGRIST & PISCHKE, *supra* note 54, at 15.

¹⁶⁶ Diana C. Mutz, Robin Pemtanle, & Philip Pham, *The Perils of Balance Testing in Experimental Design: Messy Analyses of Clean Data*, 73 AM. STATISTICIAN 32, 41 (2019).

¹⁶⁷ *Id.* at 33–34.

¹⁶⁸ Two additional participants completed the experiment but were excluded for other reasons. One participant tested the materials and had been exposed to the treatment prior to us launching the experiment. Another participant informed us that they had failed to follow

from the experiment or failed to complete it before the deadline. Ten participants in the control group dropped out of the experiment; three participants in the AI-exposed group dropped out. Ultimately, the rate of noncompletion was consistent with other studies.¹⁶⁹ Nevertheless, the disparity in attrition between the two groups raised concerns that participants in the control group may have been less likely to complete the assignment for some reason. Ultimately, we find no evidence that differential attrition affected the balance of the experiment.

The greatest threat to inference would arise if participants in the control group were more likely to drop out after reviewing the assignment because they found its requirements too burdensome. Of the thirteen participants who failed to complete the study, only two accessed the Canvas site and both participants were in the AI-exposed group.¹⁷⁰ In other words, eleven of the participants had no exposure to the assignments' details prior to dropping out. Accordingly, there is little reason to believe that participants in the control group were more likely to drop out because of the experiment's content.

Moreover, we detect no imbalance in the observable characteristics of participants who completed the experiment. Table A1 reports the distribution of observable characteristics across the two treatment groups. We find no statistically significant differences between participants in the control group

the experiment's instructions in completing the application task.

¹⁶⁹ See Michael Hoerger, *Participant Dropout as a Function of Survey Length in Internet-Mediated University Studies: Implications for Study Design and Voluntary Participation in Psychological Research*, 13 CYBERPSYCHOLOGY BEHAV. & SOC. NETWORKING 697, 699 (finding that 10% of participants drop out of a study automatically and an additional 2% of participants discontinue per 100 survey items administered).

¹⁷⁰ Both of these two participants were in the control group.

and those in the AI-exposed group.

TABLE A1: DISTRIBUTION OF OBSERVABLES ACROSS GROUPS

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	P-Value
GPA	3.39 (0.37)	3.41 (0.33)	0.749
3L Participants	0.46 (0.50)	0.46 (0.50)	0.974
AI Usage: Never	0.195 (0.40)	0.16 (0.37)	0.668
AI Usage: Last Three Months	0.39 (0.49)	0.50 (0.51)	0.299
AI Usage: Monthly	0.15 (0.358)	0.10 (0.51)	0.513
AI Usage: Weekly	0.15 (0.358)	0.20 (0.40)	0.504
AI Usage: Daily	0.12 (0.33)	0.04 (0.20)	0.169

Note: * $p < .05$; ** $p < .01$; *** $p < .001$

We find little evidence that participants who completed the experiment differ from those who dropped out. There were no statistically significant differences in GPA or class year between completers or non-completers. The distribution of responses to the question on AI usage was roughly proportional between the two groups.¹⁷¹

4. General Instructions

[Note: The following instructions were provided to all participants prior to

¹⁷¹ Given the small number of non-completers, comparing the proportion of individuals in each category proves infeasible. Five (38.5%) non-completers indicated they had used AI once or twice in the last three months. Four (30.8%) indicated they used AI once a month. Two (15.4%) non-completers indicated they used AI once a week. One (7.7%) non-completer indicated they used AI once a day. Finally, one (7.7%) non-completer indicated they had never used AI in the last three months.

the start of the study.]

1. You will need a roughly three hour block of time to complete this exercise, and the word processing software of your choice. Please complete the exercise in a private location and preserve the confidentiality of the materials. **Please complete the entire experiment in a single 3 hour setting.**

2. Please do not discuss any aspect of this exercise with anyone, both during and after the experiment.

3. Grading of your written work will focus on the quality of your analysis rather than the quality of your prose. The time limits are too short to permit the kind of nicely polished work that you would submit in practice. Focus on providing good substantive answers to the questions being asked. Substance matters far more than shine.

3. Proper Bluebooking in your written work is not important. You merely need to ensure that a reader knows what you are citing. An informal citation to a case name or Restatement provision is sufficient.

4. The target audience for your written work is a partner at your firm reading a memo rather than a law professor reading an exam.

5. The assignments in this experiment situate you in the hypothetical state of Gopher. You should assume that Gopher functions like a typical US state.

6. Each assignment has a maximum time limit, but you can finish the assigned work before that time limit. You should approach each assignment as if you are a junior attorney who is producing work for a fee-sensitive client. While you can take up to the maximum time allotment to complete each task, you should upload your memo at the point where you would feel comfortable submitting your work product to a supervising attorney, given that your client would prefer to minimize the amount they pay for your work product. If you reach the end of the maximum time and have not finished, you should upload the work you were able to produce within the allotted time. Do not spend any more than the maximum time on any assignment.

7. You will be instructed to use AI for some elements of the experiment and not others. It is essential to the integrity of the experiment that you follow these instructions.

8. Do not include your name on any of the assignments you produce for this experiment; this will facilitate blind grading, and we will be able to match your submissions with your identify through Canvas.

9. As a reminder, we will pay you \$75 upon completion of the study. Additionally, participants will have the opportunity to earn substantial

performance bonuses as follows:

- The top 40% of students will receive an additional \$50 bonus, increasing their compensation to \$125.
- The top 10% of students will receive an additional \$50 bonus on top of that, bringing their total compensation to \$175.
- You will not receive compensation if you do not complete the experiment.

APPENDIX B. SYNTHESIS TASK

1. *Synthesis Task Instructions*

[*Note:* The following instructions were provided to participants during the synthesis task.]

You have a maximum of 75 minutes to complete this assignment.

Assume that you are a law firm associate. Please write a memo in response to the following email:

We represent a client in Gopher with an interesting legal problem. I'm still figuring out the facts, but I know that the matter will turn on the doctrine of servitudes as applied to personal property (i.e. movable property/chattels), rather than real property (i.e. land). The law on this topic is complicated because most cases on servitudes involve real property rather than personal property, while our case involves personal property rather than real property.

[*Note:* The following two paragraphs were included for both groups but were highlighted for the AI-exposed group.]

Please write me a short memo (around 750 words) summarizing the current law in Gopher regarding the following issue: whether a contract for the sale of personal property that restricts how the purchaser and subsequent purchasers may use the property creates a servitude that binds subsequent purchasers who were not a party to the contract. The relevant sources on Gopher Law are in the attached source packet. Rely exclusively on these sources. Do not cite or attempt to find additional sources.

Your memo should outline the elements of the rule and any exceptions, providing a framework for how a court would approach the legal question. In other words, don't merely synthesize the sources; synthesize them into a synthesis of the rule that indicates how the elements fit together. You do not need to adhere to a traditional "memo" format. In other words, please ignore all niceties and introductions, and omit any introductory paragraph summarizing what I'm asking you to do. The synthesis can resemble an outline that clearly delineates each element of the rule.

[*Note:* Participants in the AI-exposed group also received the following separate instructions]

For this assignment, you will use the AI system Gemini 2.5 Pro to assist you in writing the assignment. Although we provide a structured process for you to follow below, you should use this process to produce your own work-product that answers the assigned question. With that in mind, please use the

following steps to use AI to assist you in producing your own work product:

Step 1: Upload the source packet. Download the source packet from Canvas. In Gemini 2.5 Pro, click the “+” button at the bottom left of the chat box. Select “Upload files” and choose the source packet you downloaded.

Step 2: Enter the shaded assignment instructions from above. Copy and paste the shaded portion of the instructions above into the Gemini 2.5 Pro chat window.

Step 3: Check AI Output. Carefully review the materials in the source packet to check the accuracy of the AI-produced output. Correct any errors. You may use AI to help with this stage if you wish.

Step 4: Refine and Edit AI Output. Refine and edit the AI output to produce a final memo that answers the assignment and that ultimately reflects your own work, although you can use text generated by AI. Again, you may use AI to help you with this stage if you wish, for example, by asking the AI to produce specific revisions. Be careful in your follow-up prompts about sources to ask only about sources in the uploaded packet.

2. Multivariate Regressions

Table B1 models quality of the synthesis task using multivariate ordinary least squares (OLS) regressions. The results are consistent with the

t-tests reported in the main text.

TABLE B1. SYNTHESIS TASK: QUALITY WITH CONTROLS

	Overall Quality	Substantive Quality	Organization	Polish
	(1)	(2)	(3)	(4)
AI-exposed	4.04*** (0.58)	1.29*** (0.22)	1.58*** (0.23)	1.58*** (0.23)
GPA	0.39 (0.83)	0.02 (0.32)	0.05 (0.33)	0.05 (0.33)
3L	-0.27 (0.59)	-0.11 (0.22)	-0.16 (0.23)	-0.16 (0.23)
Constant	6.05* (2.93)	2.12 (1.11)	2.31* (1.14)	2.31* (1.14)
AI Usage Control	Yes	Yes	Yes	Yes
N	91	91	91	91
Estimator	OLS	OLS	OLS	OLS

Note: * $p < .05$; ** $p < .01$; *** $p < .001$

Table B2 models productivity on the synthesis task using multivariate ordinary least squares (OLS) regressions. The results are consistent with the t-tests reported in the main text.

TABLE B2. SYNTHESIS TASK PERFORMANCE WITH CONTROLS

	Productivity
AI-exposed	0.09*** (0.01)
GPA	0.01 (0.02)
3L	0.01 (0.01)
Constant	0.04 (0.06)
AI Usage Control	Yes
N	90
Estimator	OLS

Note: *p < .05; ** p < .01; *** p < .001

APPENDIX C. COMPREHENSION TASK

1. *Questions*

Participants were provided the following instruction before taking the quiz:

Please answer each of the following six multiple choice questions. **YOU MAY NOT USE AI FOR THIS EXAM IN ANY WAY.** You have **ten minutes to answer these questions**. This quiz is **closed-book**. Do not consult the source packet, your memo, or anything else while taking it.

Question 1. Which of the following is NOT a potential consequence of enforcing a servitude on a chattel?

- A. Binding people to contractual terms without their express consent.
- B. Constraining the rights of the chattel's manufacturer.
- C. Replicating feudal patterns of ownership.
- D. Making the burdened chattel harder to sell.

Question 2. Suppose hypothetically that in *Parker v. Bloch* (the jukebox case), Bloch (the new owner of the bar) had wanted to add a second jukebox from a different supplier, rather than have Parker remove the existing jukebox. In this hypothetical, would Justice Jone, who dissented in the actual case, have been likely to support Parker's request for an injunction against adding the second jukebox?

- A. No, because Justice Jones believed that servitudes could never burden personal property.
- B. No, because Justice Jones favored flexibility for business owners.
- C. No, because the contract allowed Bloch to add a second jukebox from a different supplier.
- D. Yes, because in the actual case Justice Jones opposed forcing a business owner to keep an unwanted jukebox; this is not a concern in the hypothetical case, where the business owner wants jukeboxes.

Question 3. Which of the following statements about Gopher law is most accurate?

- A. Courts will invalidate servitudes burdening chattels unless the party seeking to enforce the servitude can cite closely analogous precedent supporting the servitude.

- B. Only the initial owner of property who created a servitude can enforce the servitude in court.
- C. Property owners can impose restrictive covenants on chattels that limit alienation of the chattels by subsequent purchasers.
- D. Servitudes burdening chattels are valid when an analogous servitude burdening land would be valid.

Question 4. Which of the following is NOT a benefit of enforcing servitudes on chattels?

- A. Protecting the original purchaser's expectations.
- B. Protecting the original seller's brand.
- C. Protecting the original seller's expectations.
- D. Protecting the original seller's ability to alienate the chattel.

Question 5. Which of the following statements about Gopher law is most accurate.

- A. A servitude can bind a subsequent purchaser even if the written instrument granting title to the subsequent purchaser does not mention the servitude.
- B. Affirmative covenants benefit the servient estate.
- C. Easements create possessory interests.
- D. Negative covenants are not enforceable.

Question 6. In *Norton v. Vinson* (the fruit salad case), why was trademark law arguably relevant?

- A. In *Tulk v. Moxhay* (1848), a court in England expanded the law of servitudes to protect intellectual property, and US courts found that decision persuasive.
- B. Norton had a valid trademark infringement case against Vinson.
- C. Southern Pacific and Norton had good reason for protecting Kraft's trademark.
- D. Trademarks create servitudes.

2. Question-By-Question Analysis

TABLE C: PERFORMANCE ON COMPREHENSION TASK QUESTIONS

Variable	Control Mean (Std. Dev.)	AI-exposed Mean (Std. Dev.)	Effect (% Change)	P-Value
Q1	0.69 (0.47)	0.61 (0.49)	-0.08 (-11.97%)	0.410
Q2	0.57 (0.50)	0.69 (0.47)	+0.11 (+20.10%)	0.260
Q3	0.50 (0.51)	0.71 (0.46)	+0.21 (+41.18%)	0.045*
Q4	0.50 (0.51)	0.63 (0.49)	+0.13 (+25.49%)	0.223
Q5	0.76 (0.43)	0.69 (0.47)	-0.08 (-9.93%)	0.420
Q6	0.86 (0.35)	0.57 (0.50)	-0.29 (-33.66%)	0.002**

Note: * $p < .05$; ** $p < .01$; *** $p < .001$

Table C analyzes the proportion of participants in each group that correctly answered each of the six questions. Slightly more than half of the participants answered each question correctly, suggesting that no individual question was excessively easy or unduly difficult. We observe two statistically significant differences between the two groups. The control group outperformed the AI-exposed group on Question 6, whereas the AI-exposed group outperformed the control group on Question 2. Notably, these were the only two questions that asked about the specific holdings of the named cases. We have no clear hypothesis to explain this divergence. With six independent comparisons tested at the $p < 0.05$ level, the probability of observing at least one statistically significant result by chance alone is approximately 26%, even if no true differences exist between groups.¹⁷² The two significant results we observe are therefore within the range that random variation would predict.

¹⁷² The probability that at least one test is significant is the complement of the probability that *none* are significant: $1 - 0.95^6 = 0.265$.

APPENDIX D. APPLICATION TASK

1. *Application task Instructions*

[All participants received the following instructions for the application task.]

You have a maximum of 60 minutes to complete this assignment. You are prohibited from using any AI tools.

You received the following email from the same partner who gave you the prior assignment:

Thanks for your memo on Gopher law regarding servitudes affecting personal property. I now need a memo focusing on the application of Gopher law to our client's case. We represent Craig Collins, who resides in Gopher and owns an antique car. The car is valuable because it prominently appeared in *Wheels of Justice*, a famous movie filmed in Gopher in 1945. The car is very distinctive and contains a large quantity of gold and diamonds as decorative elements.

Collins purchased the car from William Ward in 2010. Ward had previously purchased the car in 1970 from its original owner, Oscar Olson.

The car has been inoperable since the 1960s because of damage to internal mechanical components. However, the car's externally visible features are in excellent condition and retain their aesthetic beauty and original historic attributes. The sales contract between Olson and Ward contains the following provision:

"Clause 11.2. After the sale is complete, the car must not be modified. This provision is binding on Ward and all subsequent purchasers."

In a newspaper interview shortly before he died in 1971, Olson explained the rationale for Clause 11.2. Olson believed that the car had important historical significance and should be preserved as an artifact of its time and a memorial to the famous movie in which it appeared. In addition, the car was part of Olson's personal legacy: Olson was a famous actor and *Wheels of Justice* was the first movie in which he had a leading role. Olson also knew that the car brought joy to many viewers who saw it on display, and Olson wanted to preserve that opportunity for joy.

During the time that Ward owned the car, he loaned it to the Gopher Museum of History, which is the largest state-owned museum in Gopher. The museum publicly displayed the car to visitors who paid for access to the exhibit. When Collins purchased the car, he loaned it to the same museum.

The museum has continuously displayed the car since 1971. A Gopher statute enacted during Gopher's centennial celebration in 1986 declared that the car was the "official vehicle of the centennial."

Collins recently suffered a financial setback and needs to raise money; otherwise, he is likely to declare bankruptcy and will be unable to fully repay his many creditors. Collins put the car up for sale in January 2025, but there was limited demand for two reasons. First, public interest in the eighty-year-old car has diminished; it now attracts 60% fewer viewers than it did when Collins purchased it. Second, buyers are concerned that Clause 11.2 will limit their ability to earn money from the car in future years. The highest offer that Collins received for the car was \$200,000. However, a merchant contacted Collins and offered \$600,000 for the gold and diamonds in the car. The merchant did not want to buy the entire car and remove the gold and diamonds himself for fear of being sued by Olson's estate.

Collins would like to remove the gold and diamonds from the car and sell them for \$600,000. He would then install inexpensive replicas to replace the removed gold and diamonds. Removing the gold and diamonds will substantially undermine the car's historic value. Removal will also slightly undermine the car's aesthetic value because the replicas will be mostly but not entirely identical to the original items. These consequences may reduce the number of visitors who pay to see the car on display at the museum.

Collins is concerned that Olson's estate will seek an injunction preventing Collins from altering the car in this manner. Under Gopher law, Olson's estate has the same rights that Olson had before he died.

Please write me a memo that explains how a Gopher court is likely to apply Gopher's law regarding servitudes on chattels if Olson's estate seeks an injunction to prevent Collins from removing the gold and diamonds from the car. You should identify strengths and weaknesses in the client's position, recommend arguments that the client should make, and rebut counterarguments. I am looking for balanced and dispassionate analysis, not one-sided advocacy. If you need additional factual details that might influence how the court would apply Gopher law, let me know what details you need and why they might be relevant. Focus only on the law governing servitudes, and not on other potential issues in the case.

I already have your earlier memo summarizing Gopher Law, so please do not repeat or restate that memo. Instead, your assignment is to apply the law you synthesized in your earlier memo to the above facts. Doing so, of course,

will require you to mention aspects of the rule from your prior memo for your analysis to make sense, and so you should use your earlier memo to help you complete this assignment. As before, you do not need to adhere to a traditional “memo” format. In other words, please ignore all niceties and introductions, and omit any introductory paragraph summarizing what I’m asking you to do. The quality and thoroughness of your analysis matters most; to save time and ensure that you cover all relevant arguments in depth, you can outline some points rather than crafting polished paragraphs. Also as before, you should rely exclusively on the sources in the source packet rather than any other outside sources.

2. Multivariate Regressions of Performance

Table D1 models performance on the synthesis task using multivariate ordinary least squares (OLS) regressions. The results are consistent with the t-tests reported in the main text.

TABLE D1. PERFORMANCE ON APPLICATION TASK WITH CONTROLS

	Overall	Substantive	Organization	Polish
	(1)	(2)	(3)	(4)
AI-exposed	1.50** (0.52)	0.44* (0.21)	0.52** (0.19)	0.54** (0.19)
GPA	1.19 (0.74)	0.59 (0.31)	0.37 (0.27)	0.23 (0.28)
3L	-0.08 (0.52)	-0.13 (0.22)	-0.08 (0.19)	0.14 (0.19)
Constant	1.14 (2.59)	-0.11 (1.08)	0.26 (0.96)	0.99 (0.97)
AI Usage Control	Yes	Yes	Yes	Yes
N	91	91	91	91
Estimator	OLS	OLS	OLS	OLS

Note: * p < .05; ** p < .01; *** p < .001

Similar to Table D1, Table D2 models performance on the synthesis task using multivariate ordinary least squares (OLS) regressions but incorporates

participants' synthesis-task score into the model. The results are consistent with those described in the main text.

TABLE D2. PERFORMANCE ON APPLICATION TASK WITH SYNTHESIS TASK
PERFORMANCE

	Overall	Focus	Depth	Organization
	(1)	(2)	(3)	(4)
AI-exposed	0.56 (0.63)	0.25 (0.27)	0.26 (0.24)	0.05 (0.23)
Synthesis Task Score	0.23* (0.09)	0.05 (0.04)	0.06 (0.04)	0.12*** (0.03)
GPA	1.10 (0.72)	0.57 (0.31)	0.34 (0.27)	0.19 (0.26)
3L	-0.01 (0.50)	-0.12 (0.22)	-0.07 (0.19)	0.17 (0.18)
Constant	-0.26 (2.58)	-0.40 (1.10)	-0.13 (0.97)	0.26 (0.94)
AI Usage Control	Yes	Yes	Yes	Yes
N	91	91	91	91
Estimator	OLS	OLS	OLS	OLS

Note: * $p < .05$; ** $p < .01$; *** $p < .001$

APPENDIX E. REVISION TASK

1. *Instructions for the Revision Task*

[All participants received the following instructions for the application task.]

You have a maximum of 20 minutes to complete this assignment. First read the instructions below, including the instructions on how to use AI.

1. You received the following email from the same partner who gave you the prior two assignments:

Thanks for your prior two memos (i) summarizing Gopher law regarding servitudes affecting personal property, and (ii) applying that law to Collins' case. I have decided to send a final version of just your second memo to Collins, as he has asked to see it. Before I do so, please revise your second memo (the application memo) so that it is clearer and better written. Do not alter the substance of the memo, though you should feel free to elaborate on or better explain any points in the memo.

2. For this assignment, you will use the AI system Gemini 2.5 Pro to assist you in editing and improving your earlier memo. Although we provide a structured process for you to follow below, you should use this process to revise your earlier work-product as you best see fit.

Step 1: Upload your earlier memo from Assignment Three and ask for potential revisions.

- In Gemini 2.5 Pro, click the “+” button at the bottom left of the chat box.
- Select “Upload files” and choose your second memo.
- Prompt the AI to suggest potential revisions to your memo based on the Partner’s instructions above. You may do so either by copying and pasting relevant portions of the Partner’s instructions or key pieces of that instruction that you think will most improve your earlier work.

Step 2: Make revisions to your earlier memo, drawing as much as

you like on Gemini's feedback.

- You should feel free to accept all, some, or none of Gemini's comments or suggestions. Make sure not to use any suggestions that draw from new sources outside of the source pack.

Step 3: Consider using Gemini AI to improve individual sentences or paragraphs.

- As you see fit, you may ask Gemini to improve individual sentences or paragraphs within your memo. You can do so simply by prompting Gemini to "Improve this" and then cutting and pasting the revisions you like. As always, make sure that any incorporation of Gemini's suggestions faithfully reflects your intended meaning.